

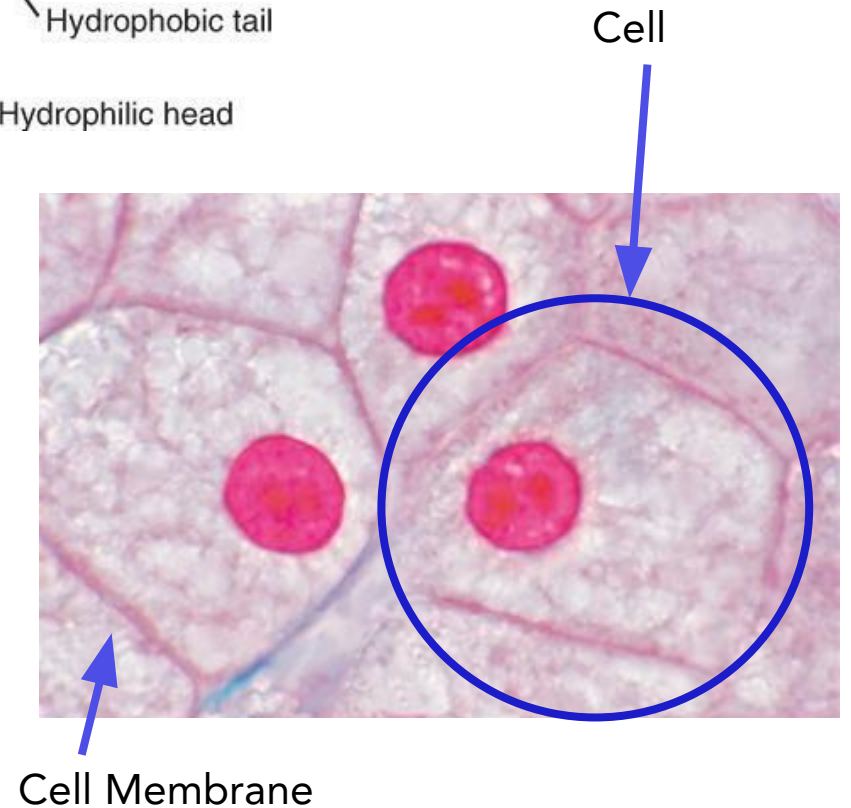
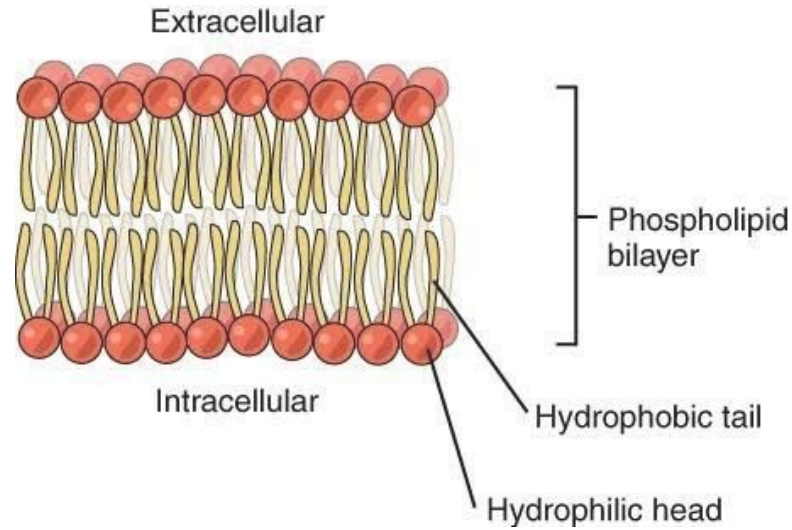
A microscopic image of plant cells, likely onion skin, showing cell walls and large central vacuoles. The image is overlaid with a semi-transparent purple circle on the left side, which contains the text. The cells are stained, with the vacuoles appearing pinkish-red and the cytoplasm and cell walls appearing blueish-purple.

Unit 2 (Part 1)

Cell Membrane and Transport

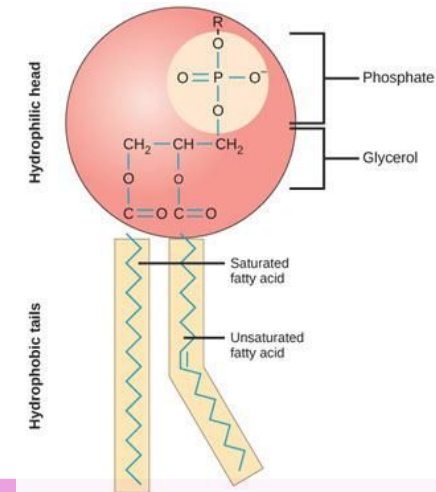
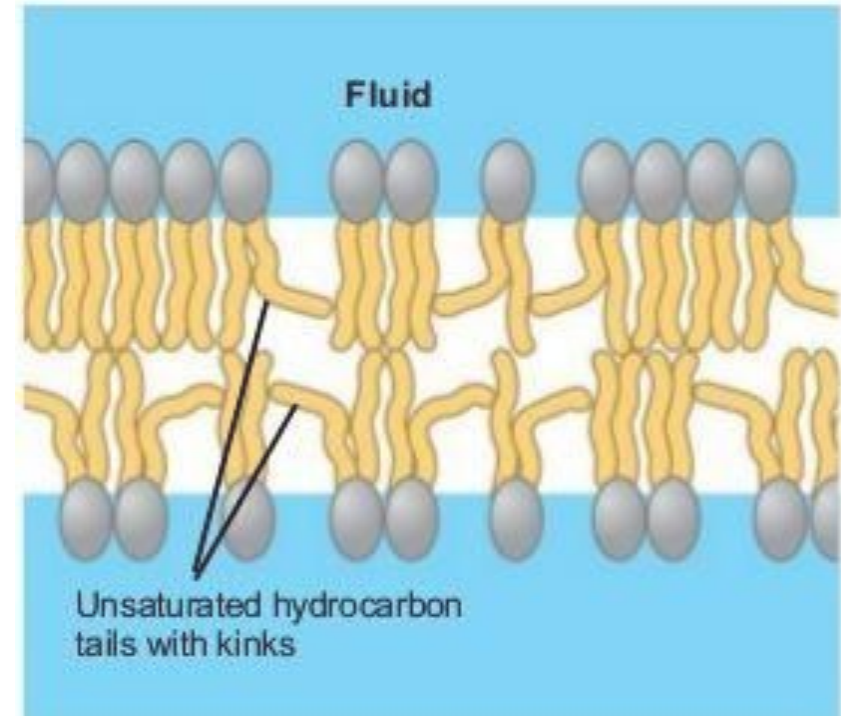
Cell Membrane

- **cell** - smallest functional unit of organisms
- **cell membrane** - regulates entry and exit of molecules
 - AKA plasma membrane or phospholipid bilayer (2 layers of phospholipids)
 - surrounds **ALL cells**



The Membrane is Fluid & Flexible

- plasma membrane's **fluidity** and **flexibility** comes from:
 - Unsaturated fatty acid chains' double bonds cause bends and allow them to move freely)
 - Length of fatty acid chains makes it soft and flexible

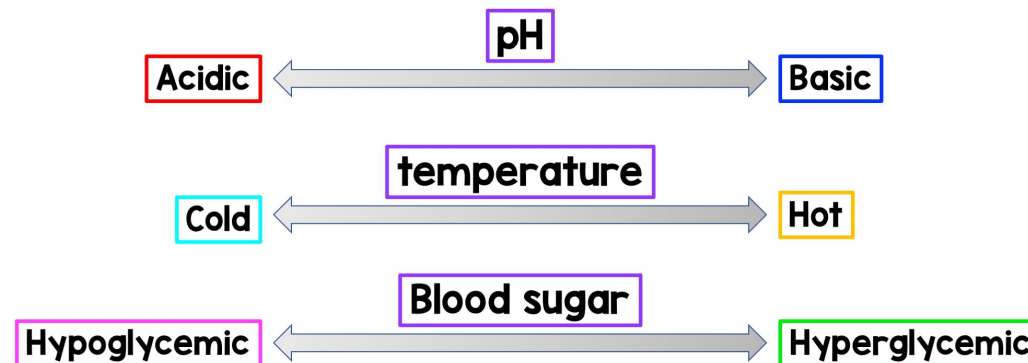


Cell Membrane Functions

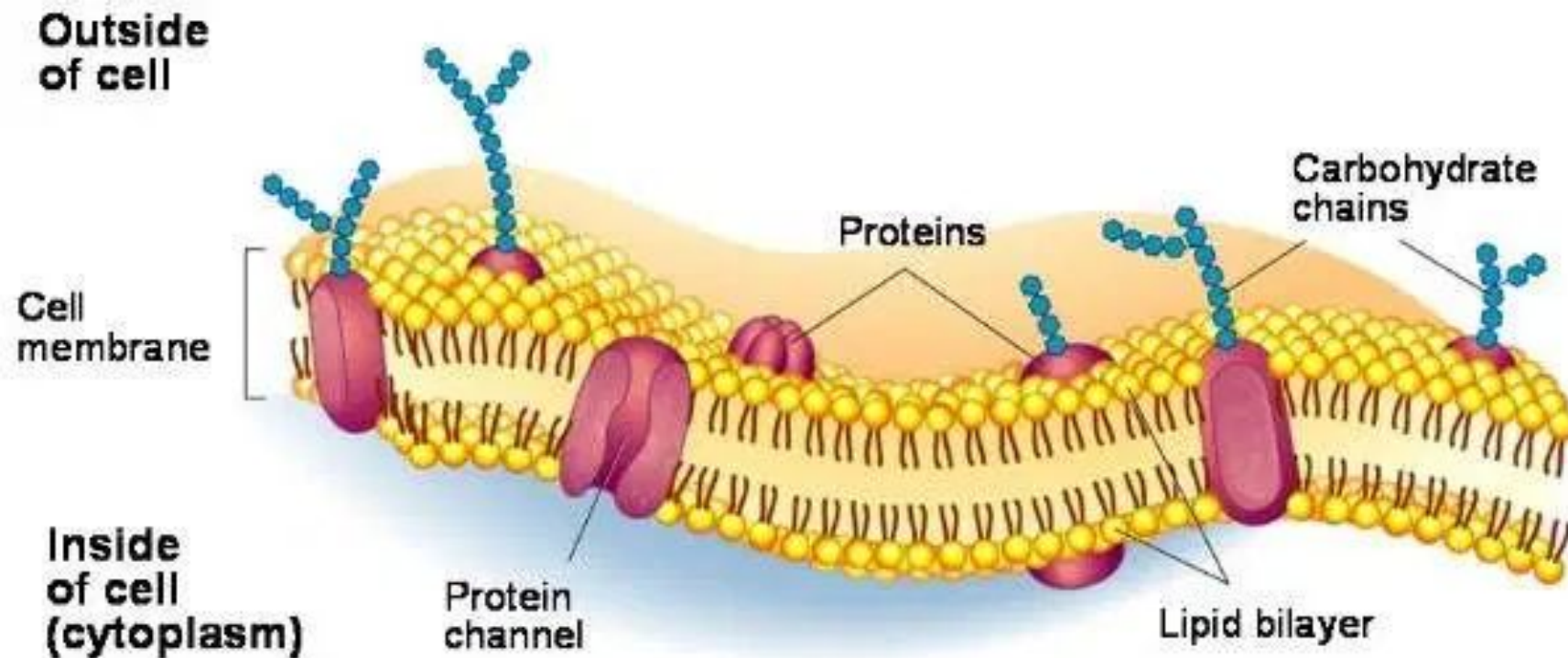
1. Contains cytoplasm and all cell parts
2. Protective barrier between inside and outside of cell
3. Regulates things coming in and out of the cell
4. Anchor for cytoskeleton
5. Provides a binding site for enzymes
6. Interlocking surfaces ("junctions") allow nearby cells to bind together for cell-to-cell recognition

Homeostasis

- **homeostasis**- maintaining/regulating a **stable internal environment** despite changes in external conditions
- maintained by the **cell membrane regulating entry and exit**
- ensures cells have the correct amount of everything (nutrients, water, proteins, etc.)

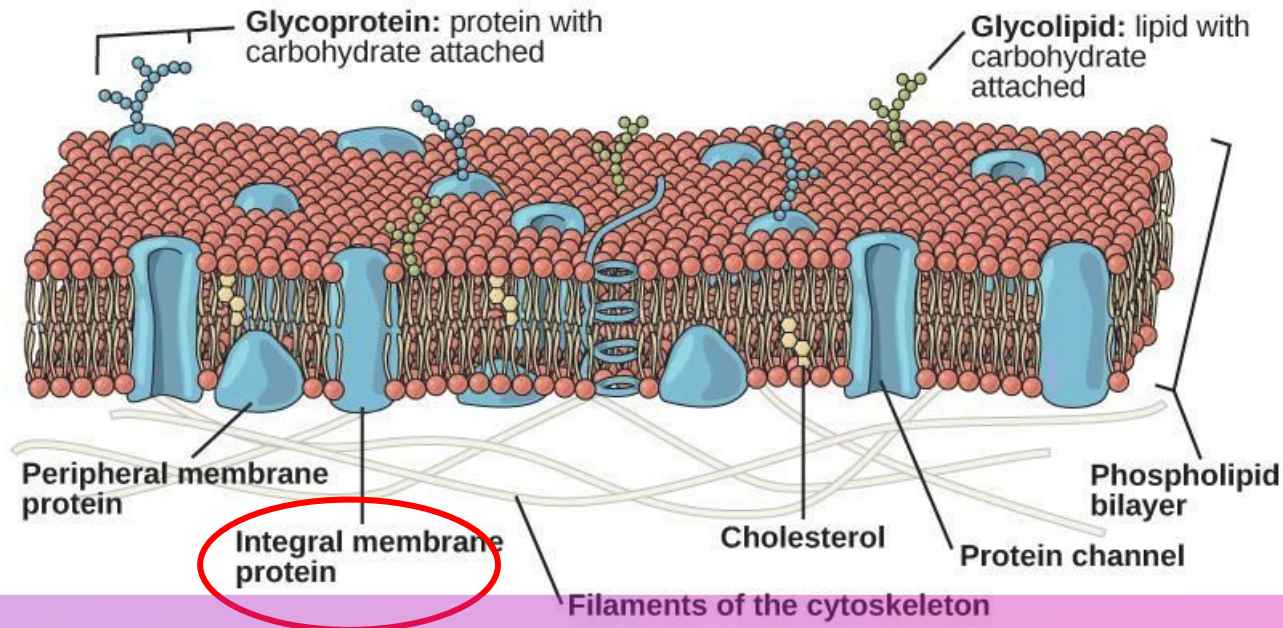


Macromolecules in the Cell Membrane?



Integral Proteins

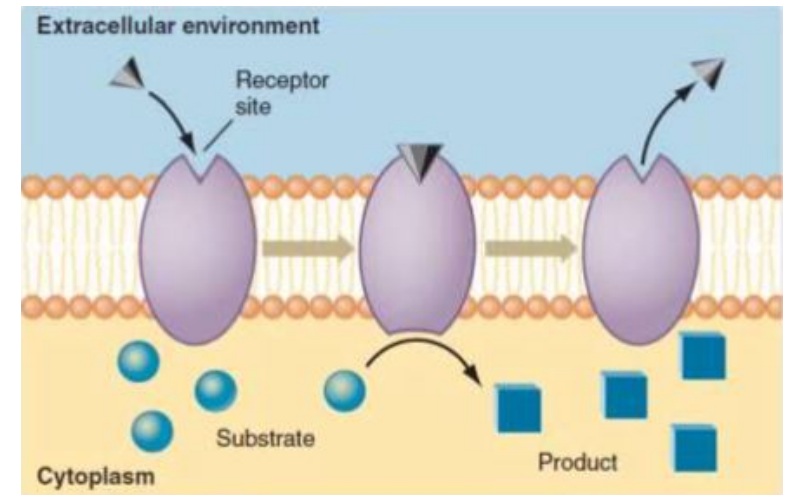
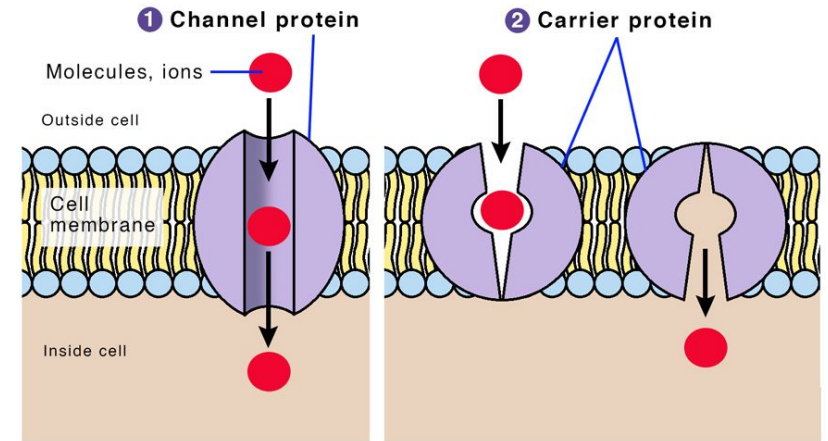
- **integral proteins**- span (go across) the whole membrane from one phospholipid head to another
- help the movement of large molecules that cannot simply pass through



3 Types of Integral Proteins

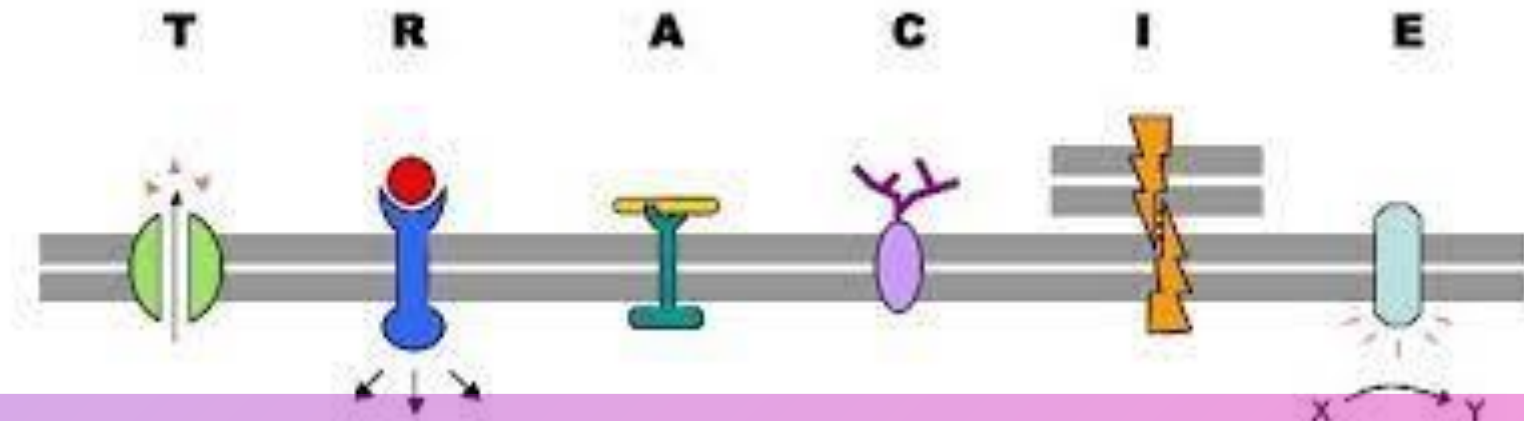
1. **carrier protein**- a type of **transport protein**; changes shape to let water molecules or small ions through quickly
2. **channel protein**- type of **transport protein** with tunnel-like structure in the middle allows molecules to pass through
3. **receptor protein**- binds to signaling molecules (like hormones)

Types of Transport Proteins



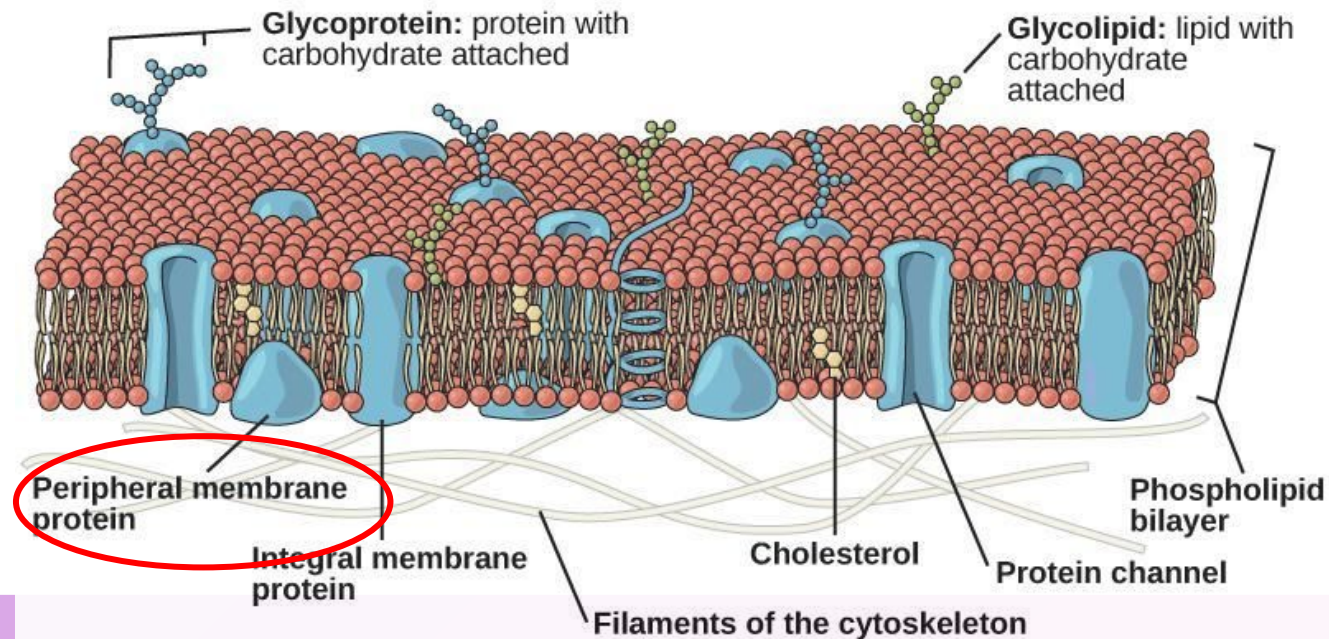
Membrane Protein Jobs (TRACIE)

- **T**ransport: channels and pumps
- **R**eceptors: hormones (ex. insulin)
- **A**nchorage: Cytoskeleton attachment
- **C**ell **R**ecognition: Immunity
- **I**ntercellular **J**oining: Tight junctions and plasmodesmata
- **E**nzyme **A**ctivity: Speed up chemical reactions



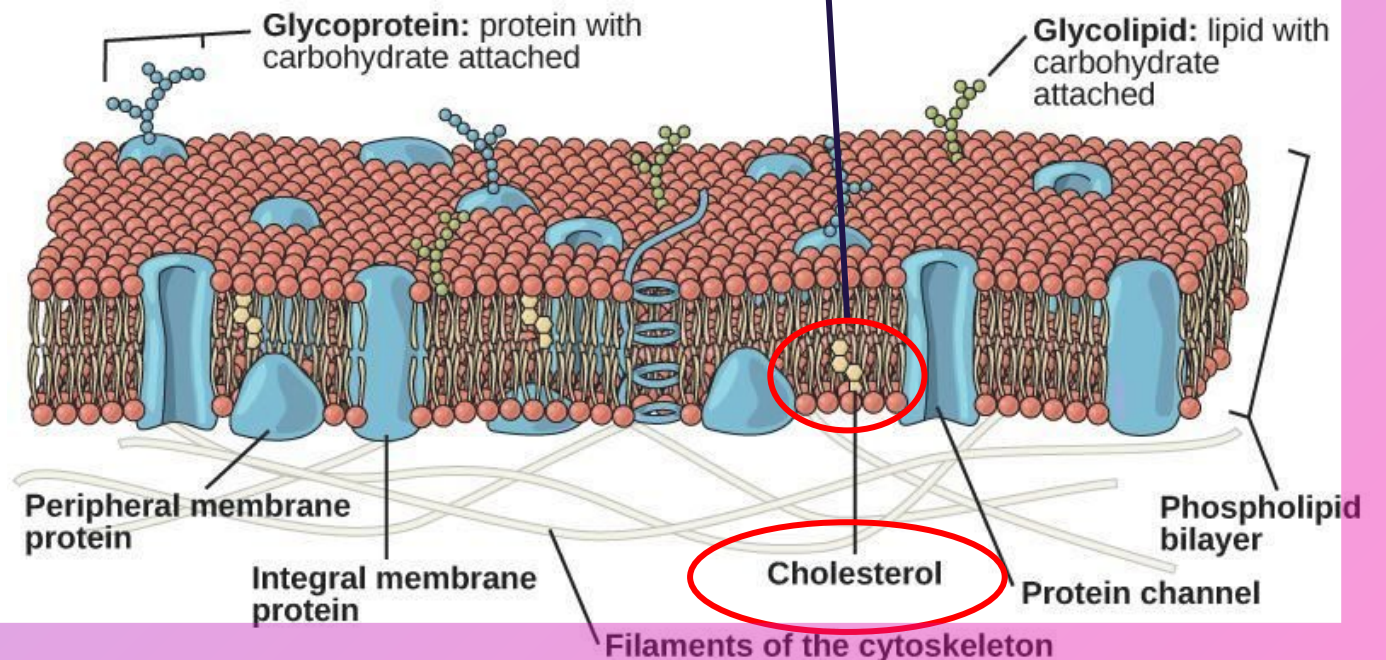
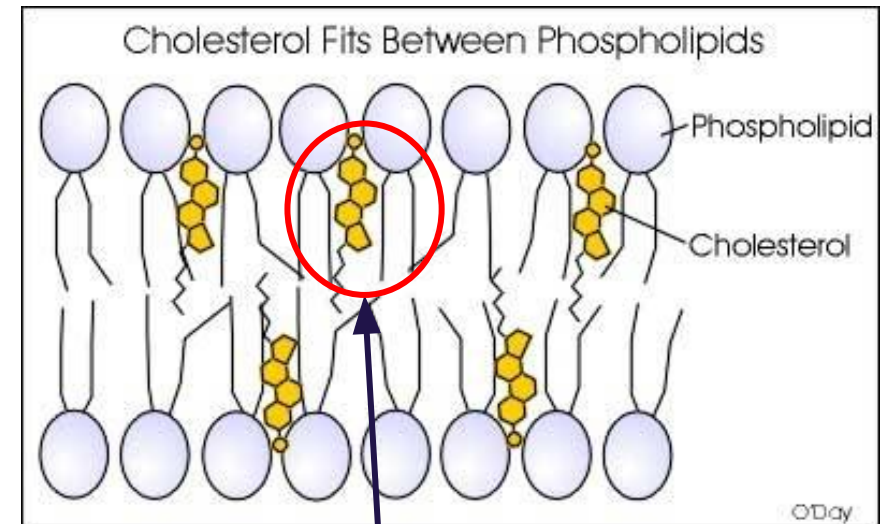
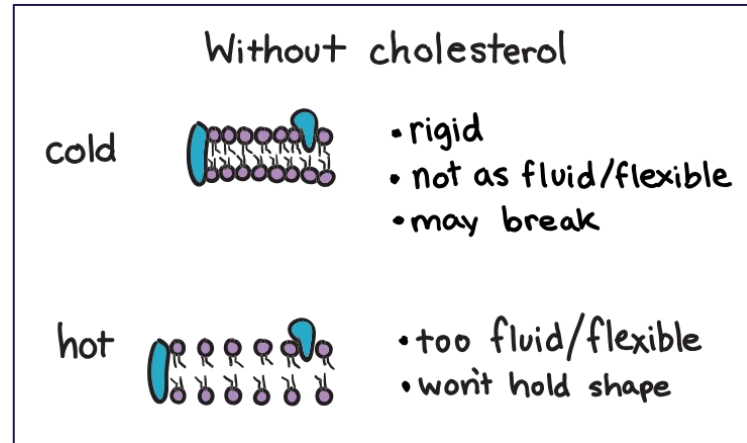
Peripheral Proteins

- **peripheral proteins**- located on **interior or exterior surface of membrane (edges)**
- **cell recognition** (“I know you, fellow cell.” or “I don’t know you; you could hurt me.”)
- **hormone recognition** (“This is a message from somewhere.”)



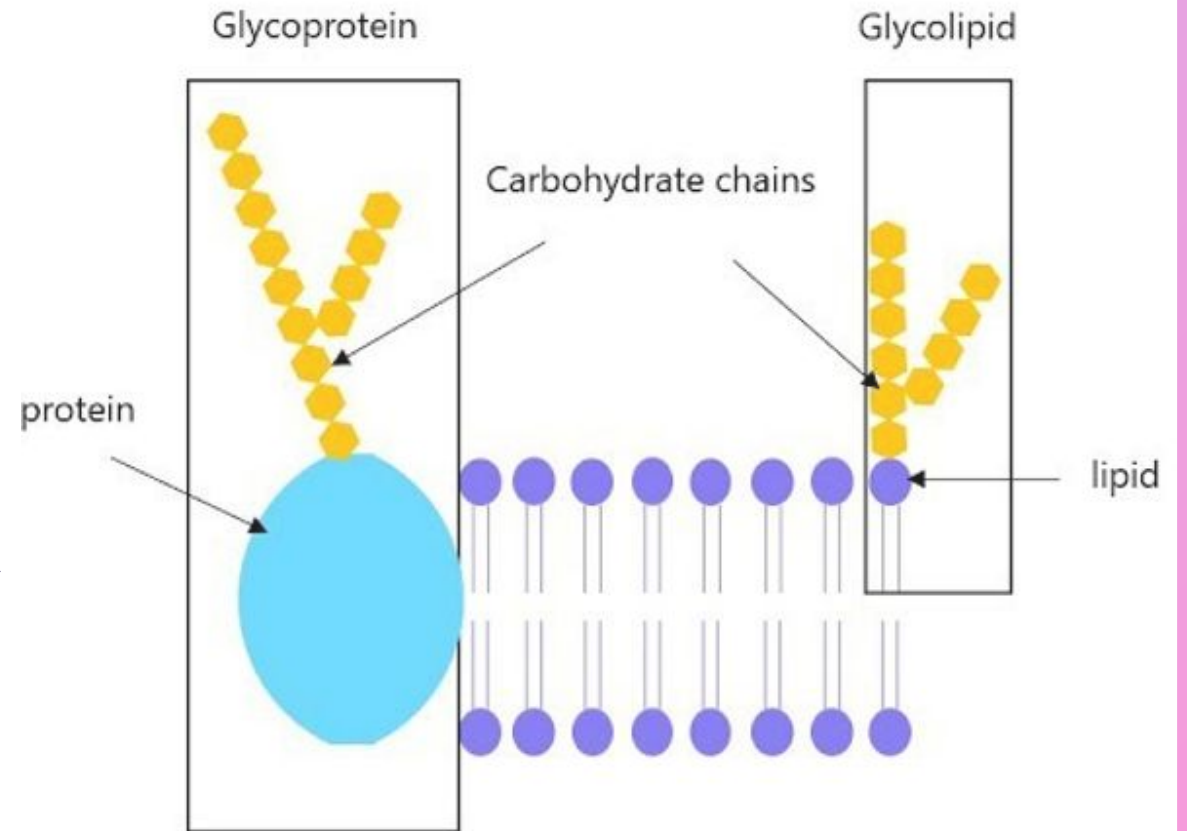
Cholesterol

- cholesterol- steroid (4-fused C ring structure) that keeps cell membrane from being too solid or fluid
- provides stability to membrane by physically limiting movement of the phospholipids
- acts as a temperature buffer

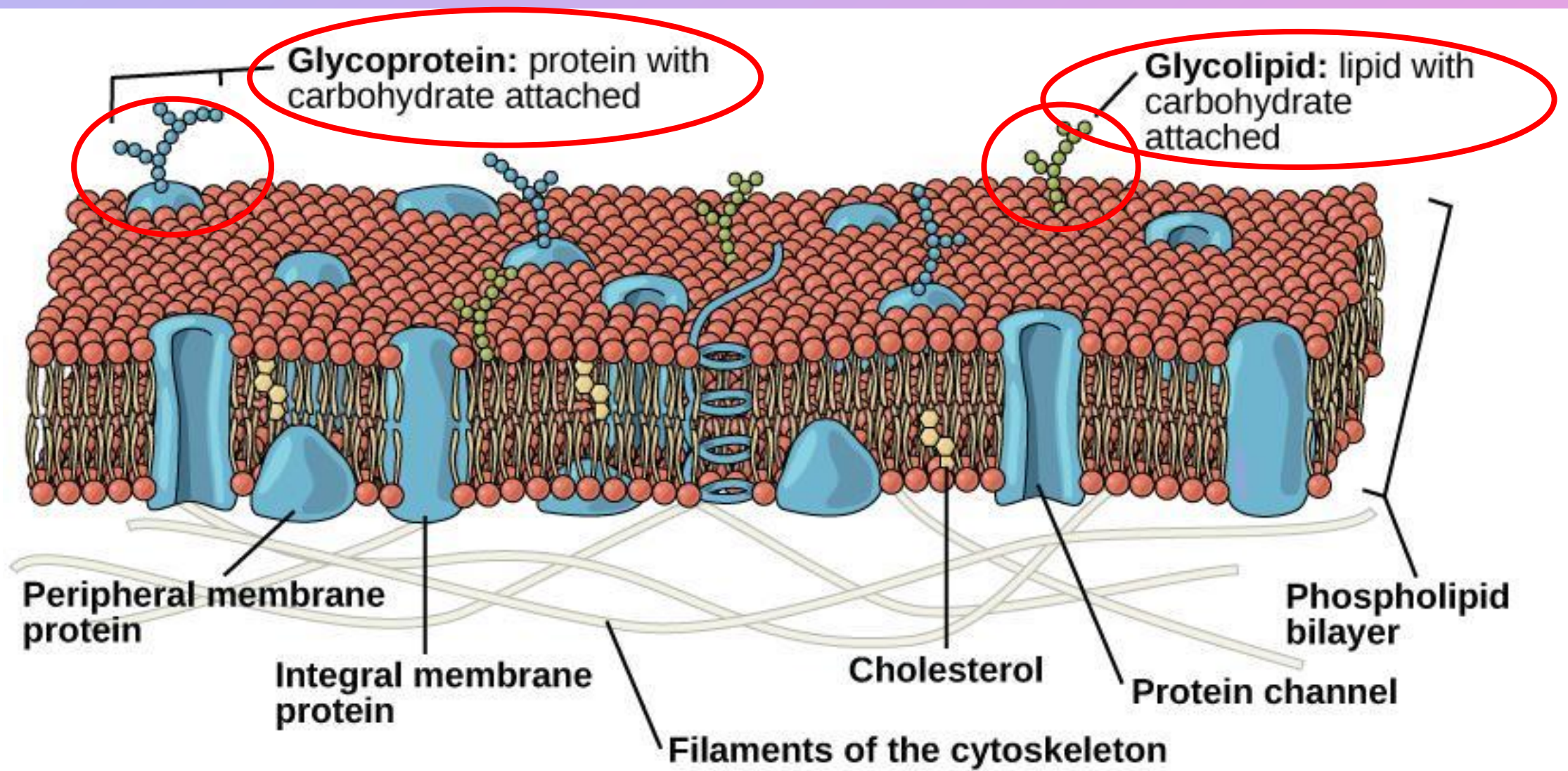


Oligosaccharides

- **oligosaccharides**- linked chain of a few (“oligo-”) monosaccharides that allow for **cell-to-cell recognition**
- **glycolipids**- **carbs** that are attached to **lipids** within the bilayer
- **glycoproteins**- **carbs** that are attached to **proteins** (can be integral or peripheral)
- provides stability to the membrane
- important part of the immune system (**WHY?**)

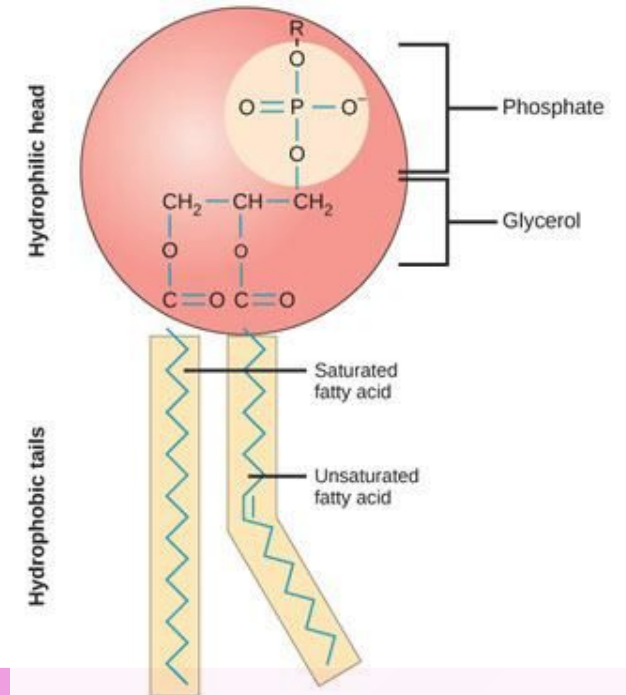
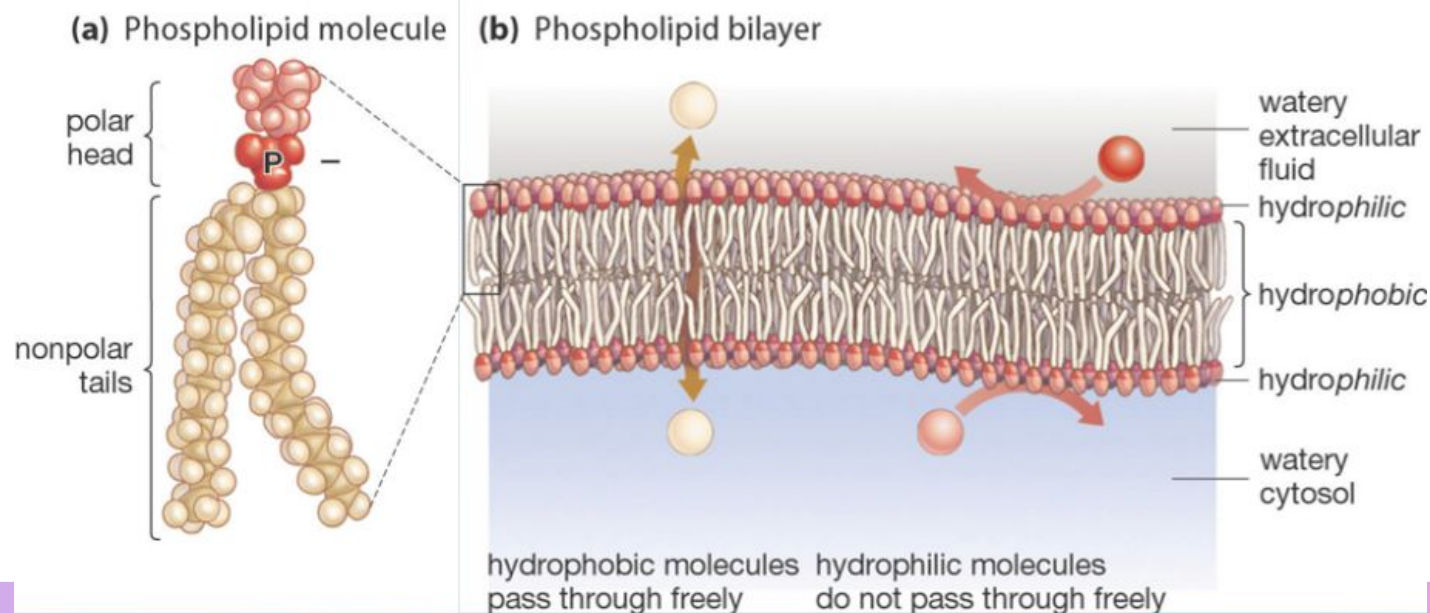


They're like tiny little name tags!



Selective Permeability

- **amphipathic**- having both hydrophobic AND hydrophilic properties
- make the cell membrane “selective,” only letting certain substances through
- polar heads are hydrophilic; nonpolar tails are hydrophobic



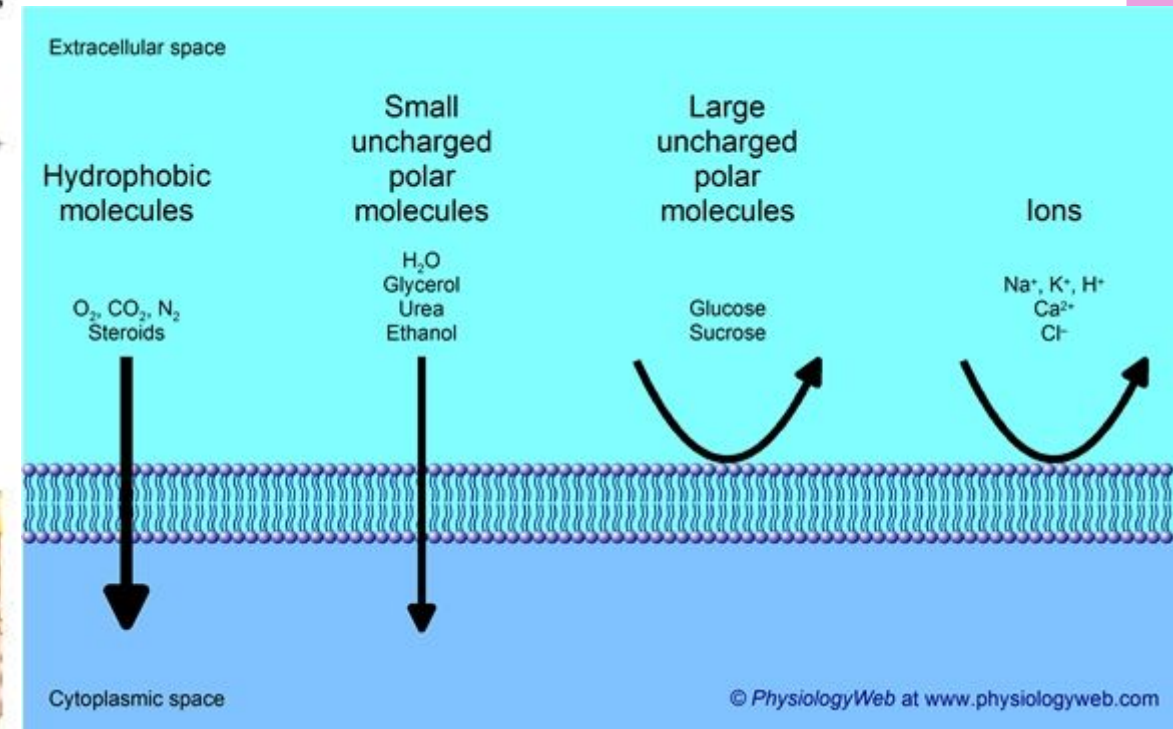
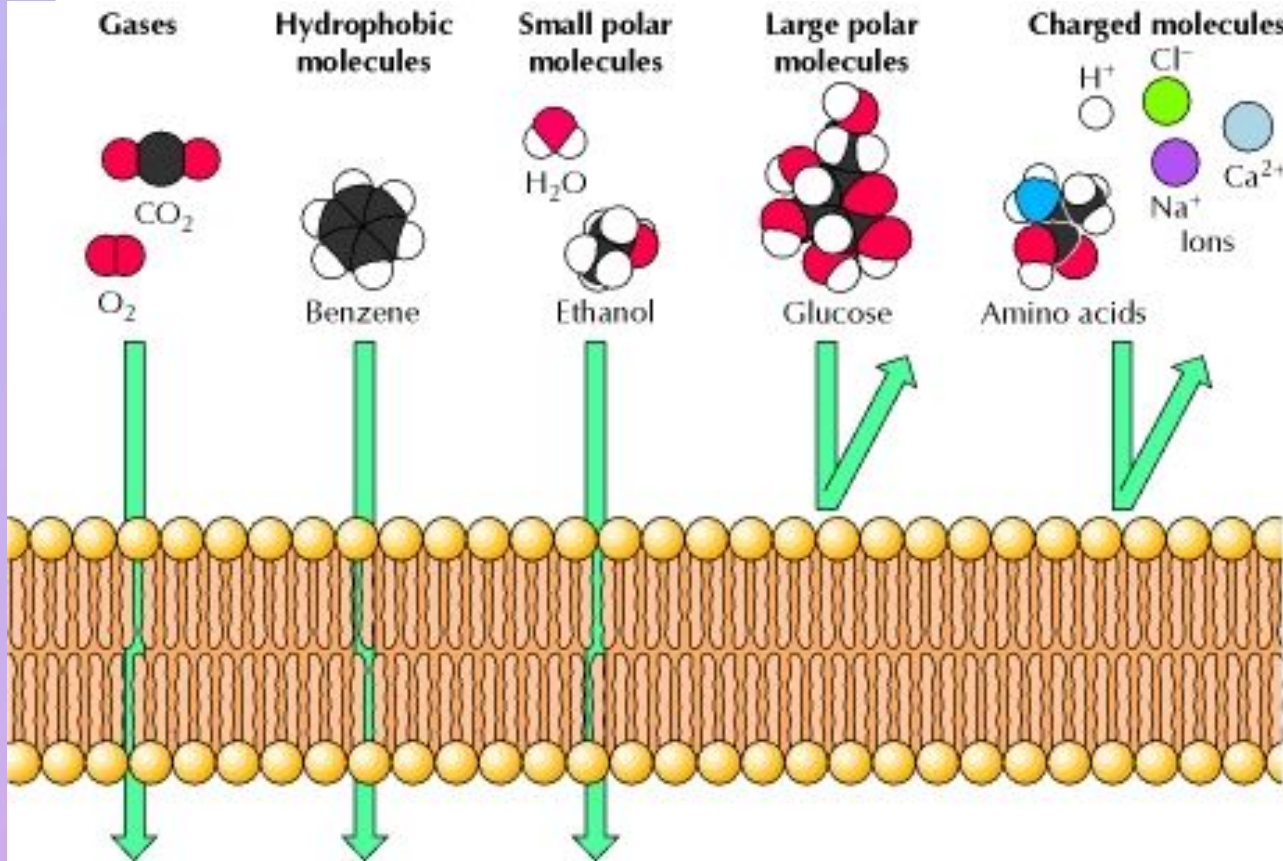
Molecules that DO Pass Easily

- Small, uncharged polar molecules (ex. H_2O)
- Gases (ex. O_2 and CO_2)
- Lipid-soluble molecules
- Large hydrophobic molecules

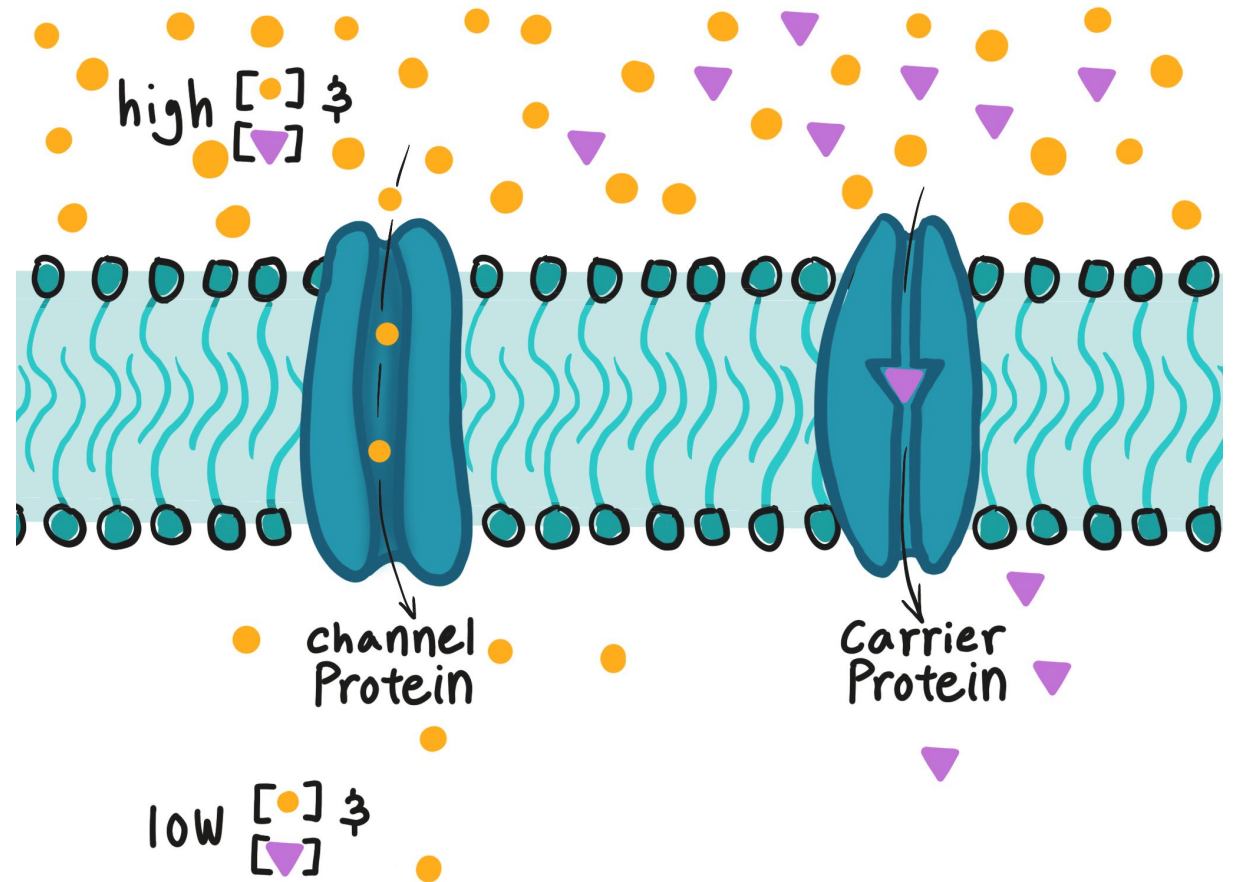
Molecules that DO NOT Pass Easily

- Ions (have a + or - charge)
- Hydrophilic molecules LARGER than water (ex. glucose, $\text{C}_6\text{H}_{12}\text{O}_6$)
- Large molecules that cannot fit or move on their own (ex. proteins)

Molecules that DO and DO NOT Pass Easily

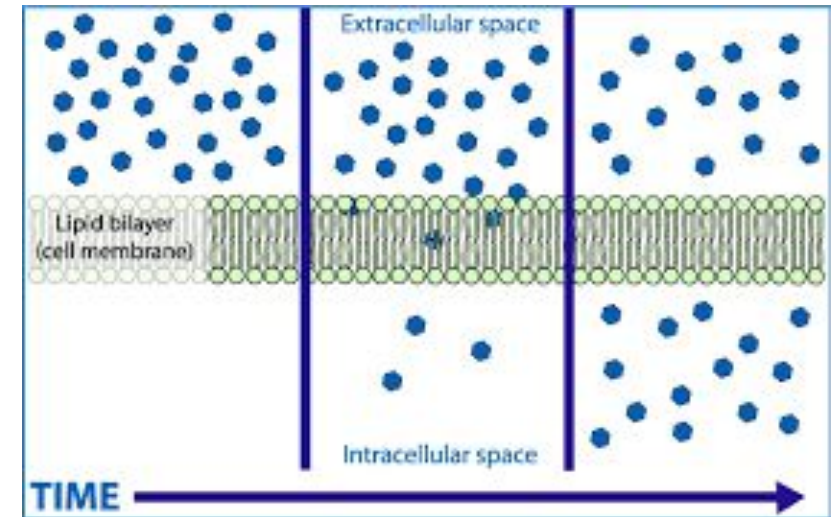


Passive Cell Transport



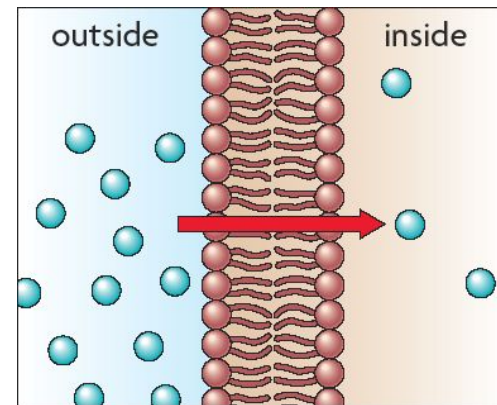
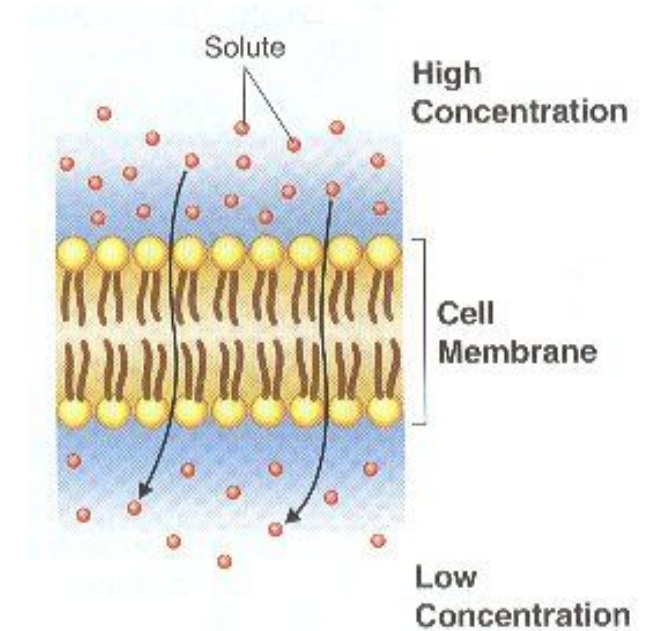
Passive Cell Transport

- **Equilibrium**- equal concentration of solutes on both sides of the membrane
- Cells want to reach equilibrium
- **Passive transport**- no energy required
 - Molecules move down the concentration gradient □ from area with a HIGH concentration of solutes TO an area of LOW concentration
 - Must have a concentration gradient (unequal amounts of solute on each side) to occur



Passive Transport: Simple Diffusion

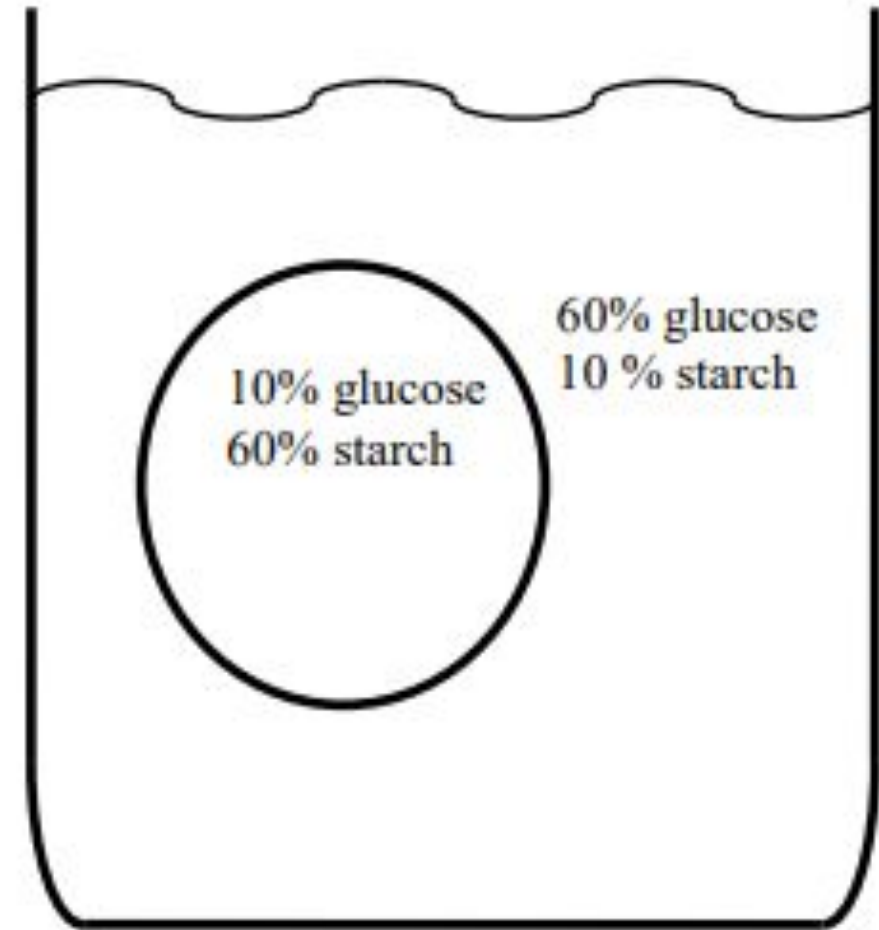
- **Simple diffusion**- movement of molecules (fluid or gas) from an area of **HIGH to LOW** concentration until equilibrium is reached
- Ex. **Oxygen, carbon dioxide and other small molecules**



Practice: Diffusion

The diagram below shows a cell in a solution. **The cell's membrane is permeable to starch only.**

1. Which molecule will move?
2. What direction will it move?



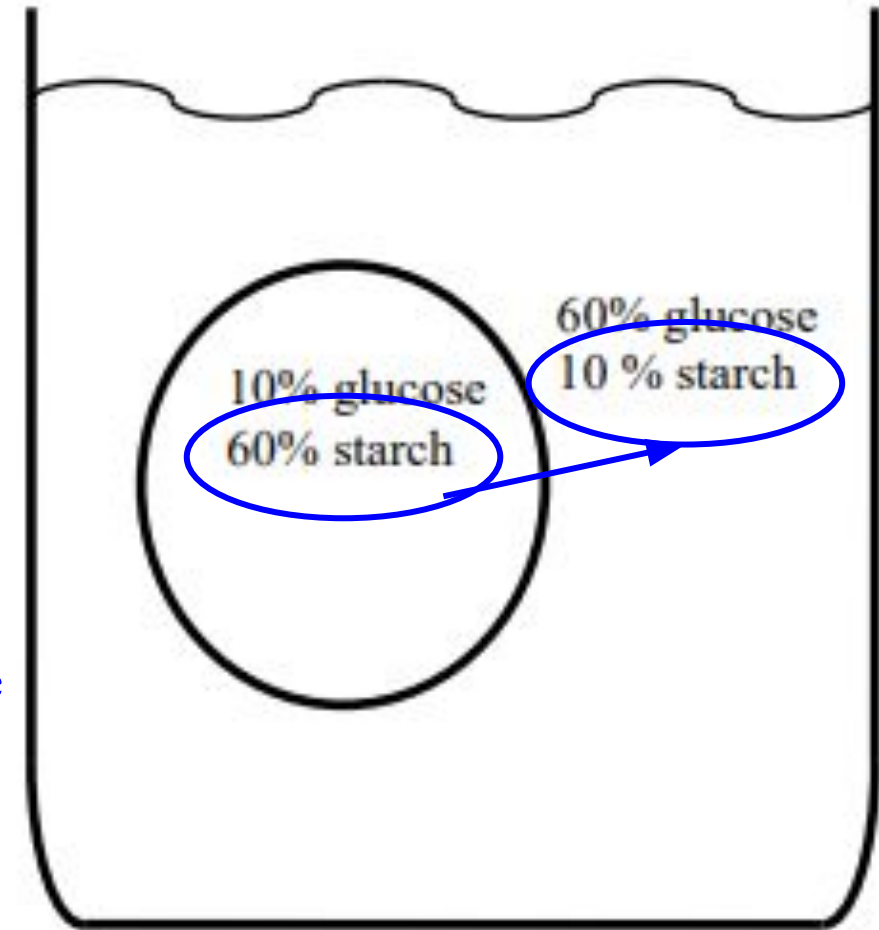
Practice: Diffusion

The diagram below shows a cell in a solution. **The cell's membrane is permeable to starch only.**

1. Which molecule will move?
2. What direction will it move?

1. **Starch only.** The membrane is permeable to starch **ONLY**, not glucose, meaning that only starch can move through.

2. **Diffusion is one of the 3 types of passive transport.** Passive transport does **NOT** require energy (ATP.) Molecules will move from **HIGH** → **LOW** concentration (down, or with, the concentration gradient). Starch will move from high (60% inside) to low (10% outside). Starch will move **OUT** of the cell.

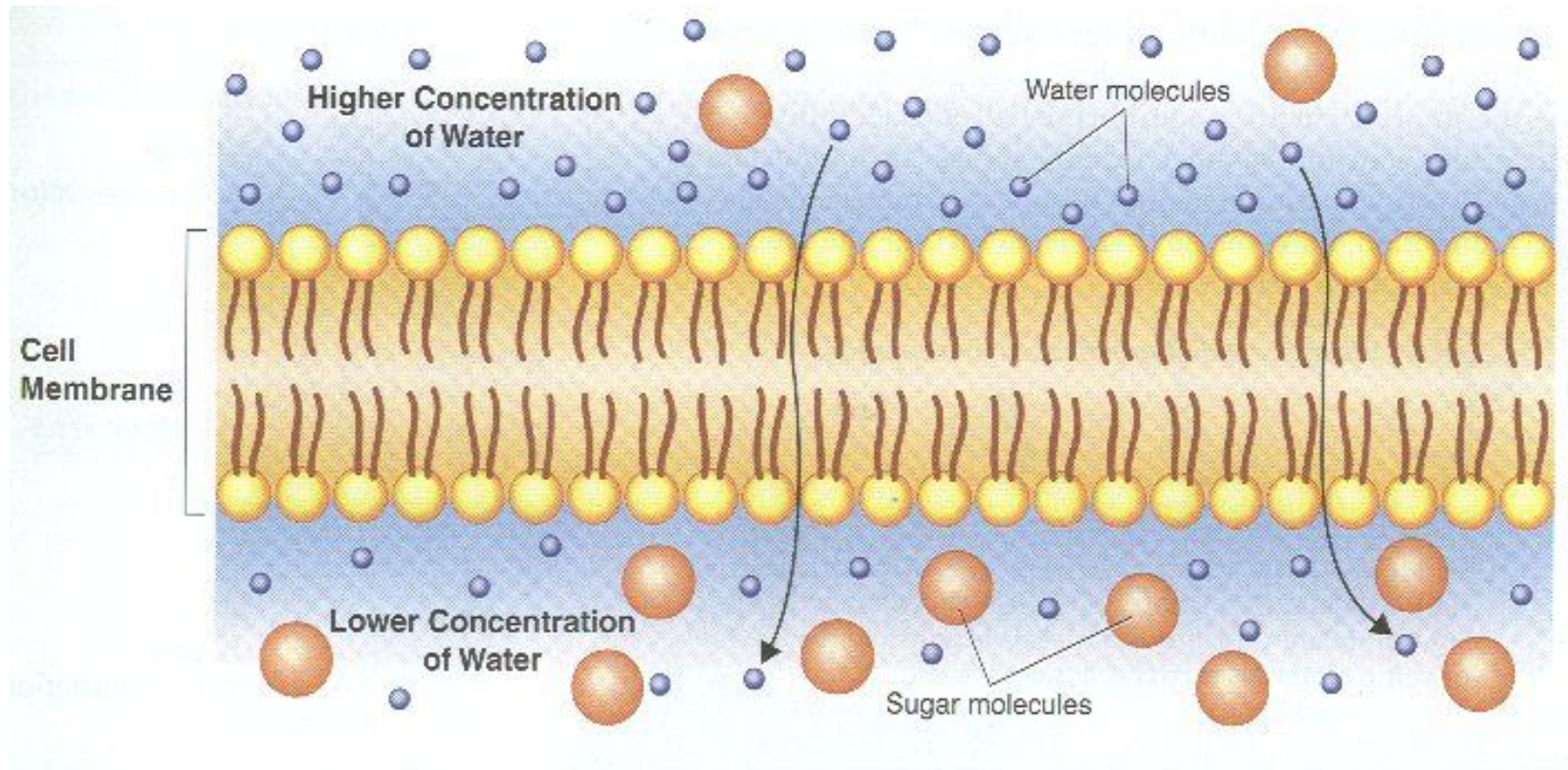


Passive Transport: Osmosis



- Osmosis- simple diffusion of water across the cell membrane
- Water molecules move down a concentration gradient
 - □ from an area of HIGH water potential, to LOW water potential
 - High water potential = low solute concentration
 - Low water potential = high solute concentration
 - “Water slide” → water moves from HIGH → LOW, doesn’t require energy (you can slide down a water slide from high to low without much energy)

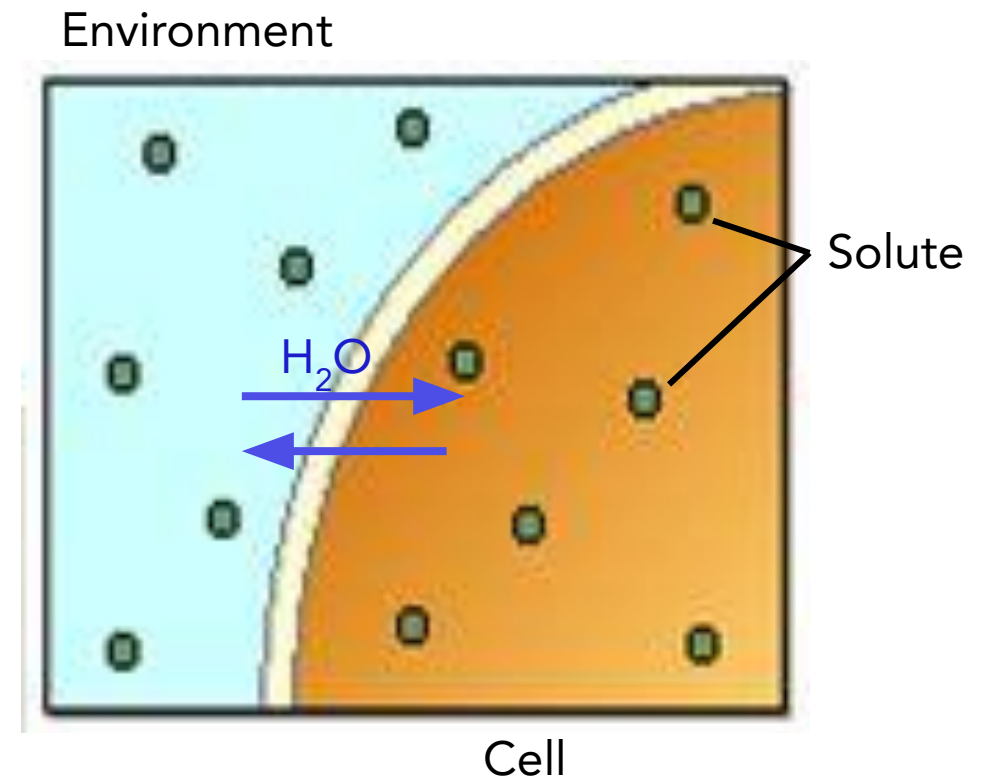
Osmosis



Water is moving towards solutes from HIGH water concentration → LOW water concentration

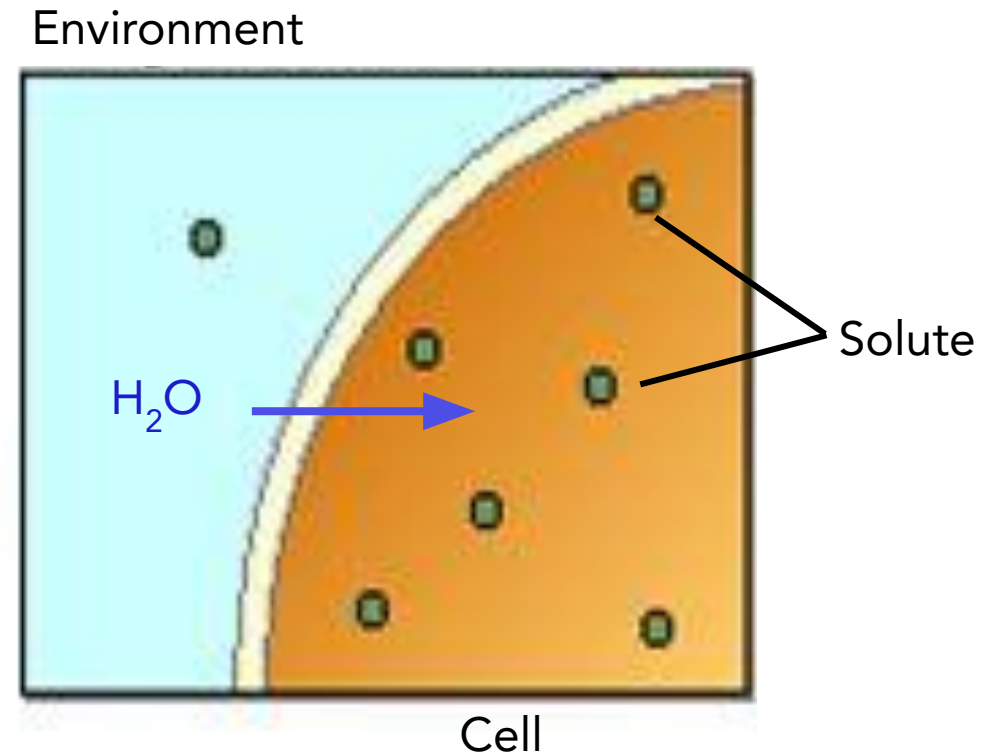
Osmosis: Isotonic Solution

- **Isotonic**- equal concentrations of **solute** in and out of the cell
- Water will move both **IN AND OUT** of the cell at an equal rate
- Cell is at **dynamic equilibrium**



Osmosis: Hypotonic Solutions

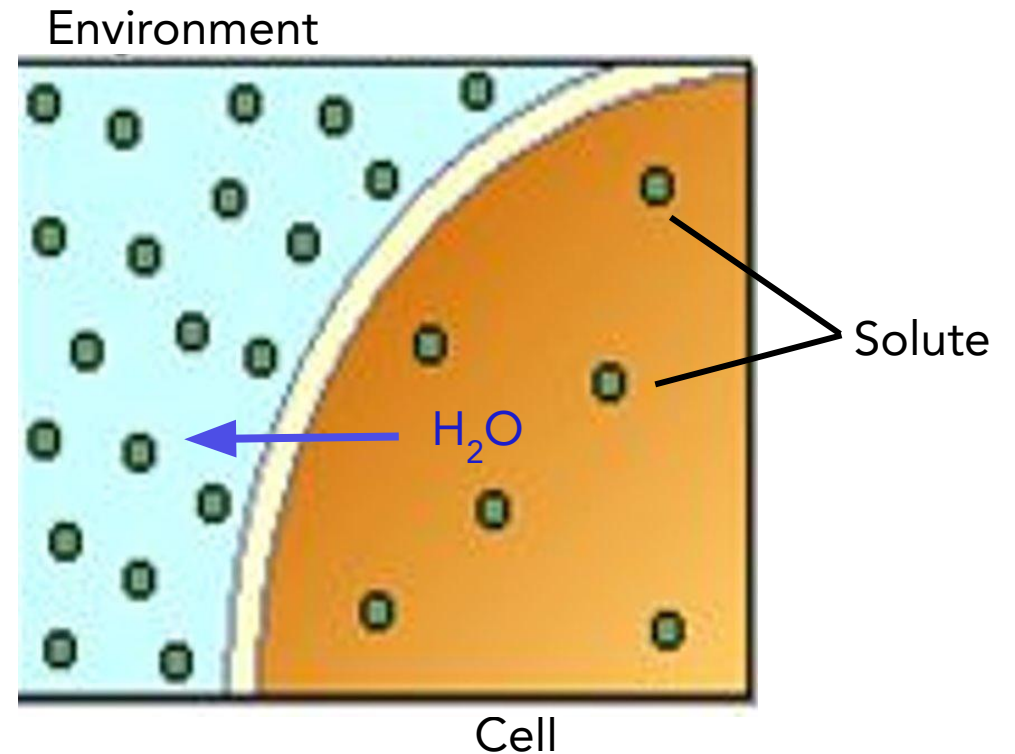
- **Hypotonic solution**- solute concentration is **lower outside** of the cell than inside of the cell
- **Water will rush INTO** the cell; cell swells like a **hippo**
- “**hypothermia**” = **low** temperature ::
“**hypotonic**” = **low** solute



Osmosis: Hypertonic Solution

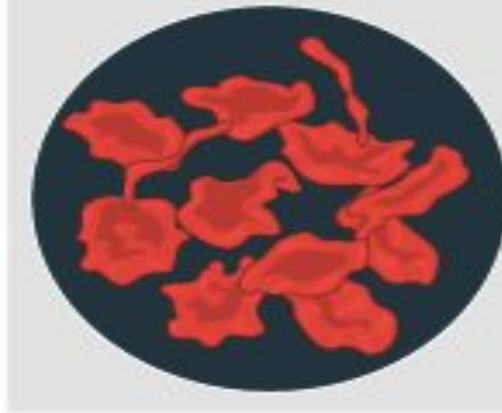
- Hypertonic solution- solute concentration is **higher outside** of the cell
- water will move **OUT** of the cell; cell will shrivel up/shrink

When you are “hyper,” you sweat and lose water.



Osmosis in Animal Cells

**Hypertonic
Solution**



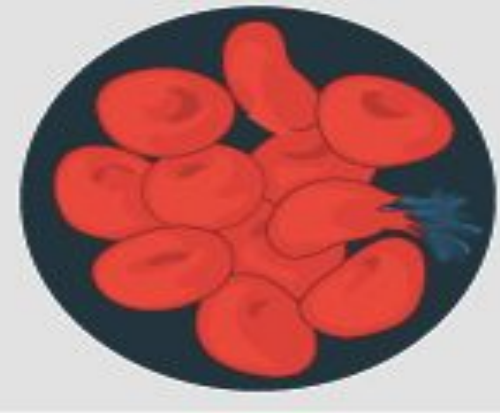
Cell **shrivels** (**plasmolysis**) =
Dead/shriveled animal cell 😞

**Isotonic
Solution**



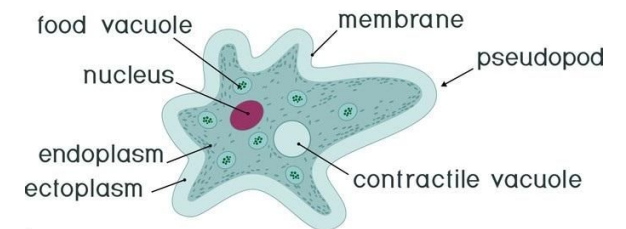
No net H₂O movement =
Happy animal cell 😊

**Hypotonic
Solution**



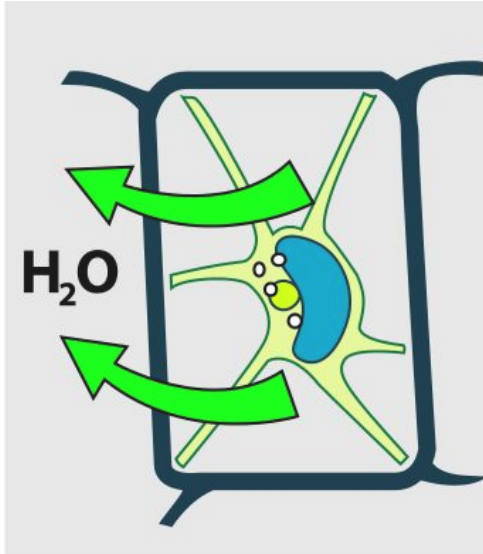
Cell swells and **bursts** (**cytolysis**) =
Dead/bursted animal cell 😞

****Contractile vacuoles-** Not found in animal cells, but are found in **protists** that live in freshwater (like Amoeba). These organelles help push out excess water & help prevent cytolysis in these organisms' cells.



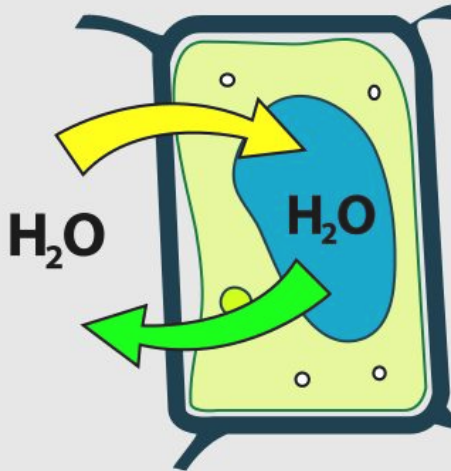
Osmosis in Plant Cells

**Hypertonic
Solution**



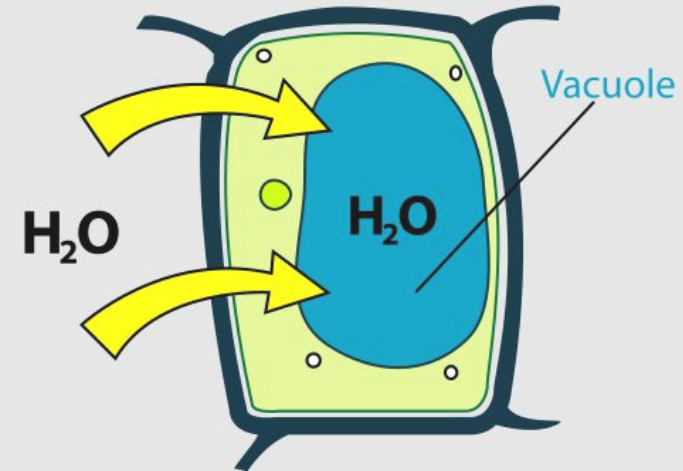
Cell **shrivels** (**plasmolysis**) =
Dead/wilted plant cell 😞

**Isotonic
Solution**



No net H₂O movement =
Alive, but limp plant cell 😐

**Hypotonic
Solution**



Cell is **turgid**** =
Alive, **happy plant cell** 😊

****Turgor pressure**- force within a cell caused by fluid that pushes the cell membrane against the cell wall. Plants can hold more water & resist bursting more than animal cells due to a **large central vacuole** & **cell walls**. A plant is described as **turgid** when it is slightly swelled and well-watered.

Practice: Osmosis

The beaker to the right is divided in half by a semipermeable membrane. Both sides are filled with water and **sucrose molecules (red)**. The membrane is permeable to water only.

1. Which side has the highest water potential?

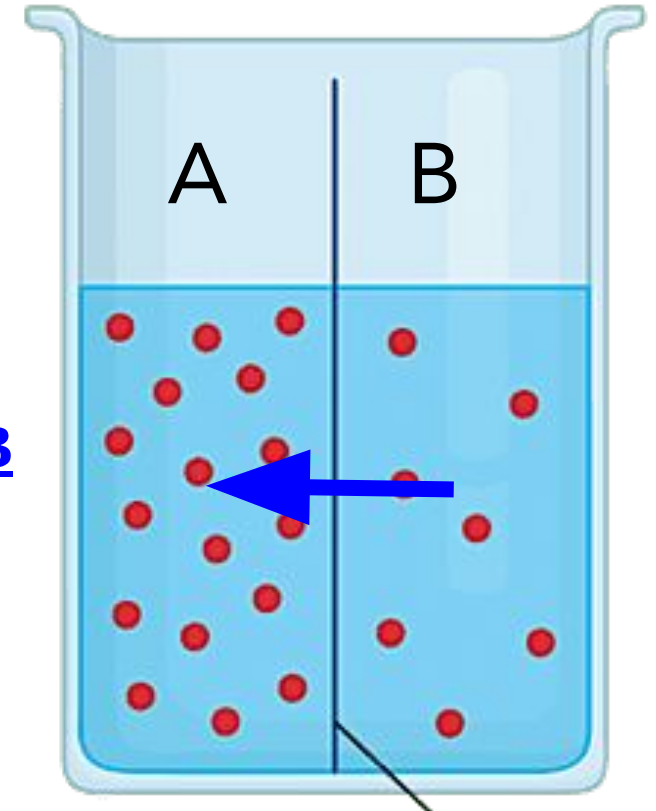
“Highest water potential” = “highest water concentration” → **B**

2. Which side has the highest solute concentration?

A (more red sucrose molecules in the same volume of water)

3. In which direction will water move?

Osmosis is the simple diffusion of water specifically. Diffusion is passive transport and requires no energy (ATP). Water will move from where there is HIGH water (B) → LOW water (A). Water will move from B → A (left).



Practice- Osmosis

- The cell to the right is placed in a solution of glucose. The cell membrane is only permeable to water.

1. What type of **solution** surrounds the cell?

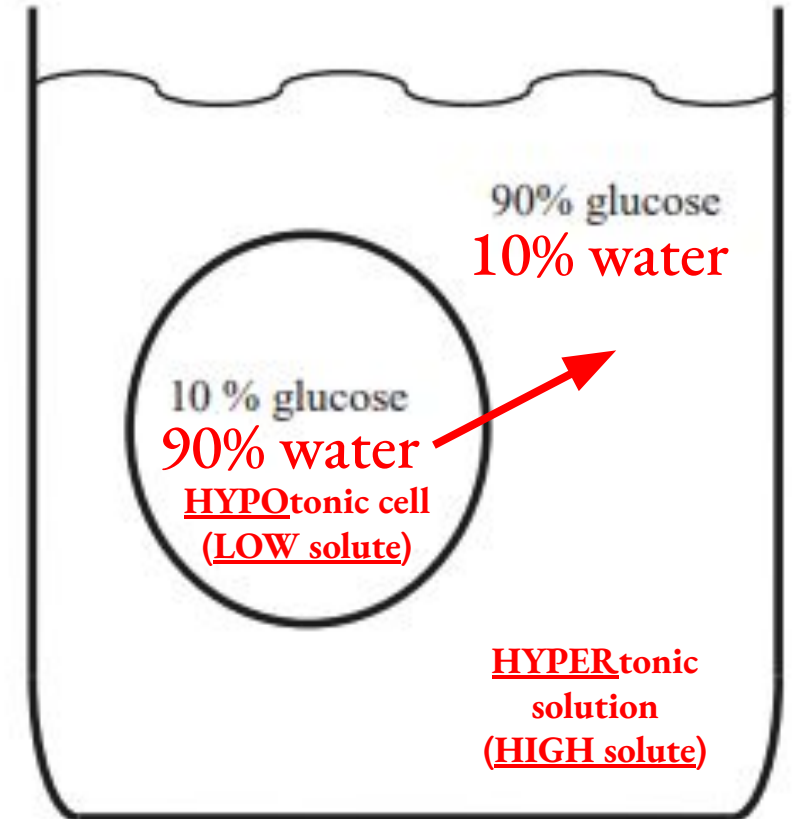
The cell is surrounded by a **HYPERTONIC SOLUTION** (**HYPER**tonic = **HIGH SOLUTE** concentration).

2. Where will the water move?

Water will diffuse (move) from high → low water concentration **OUT** of the cell.

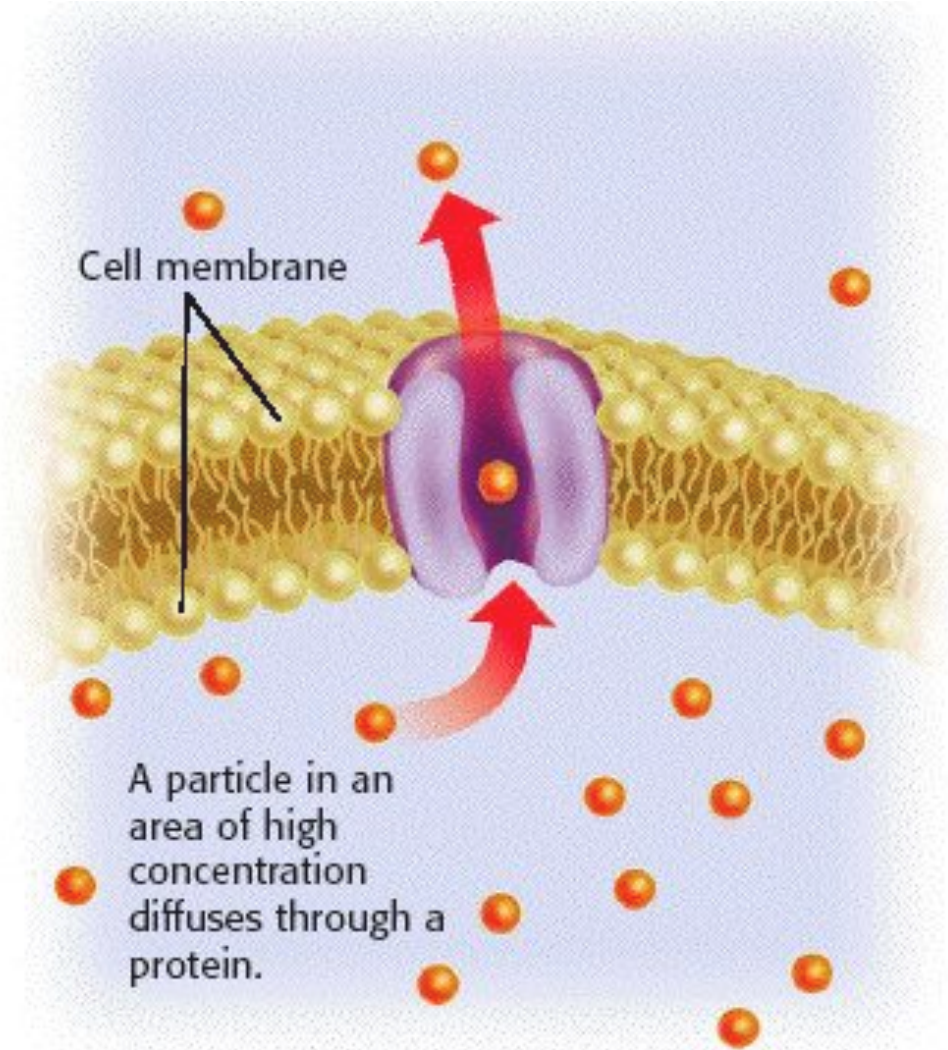
3. What will happen to the cell after the water moves?

The cell will **shrink/shrivel** as water moves out. (**plasmolysis**)



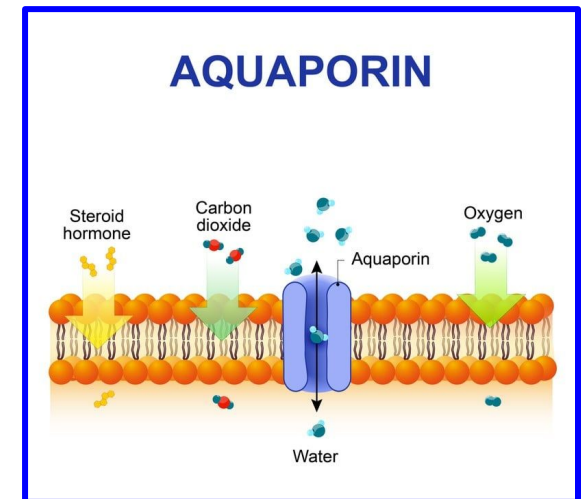
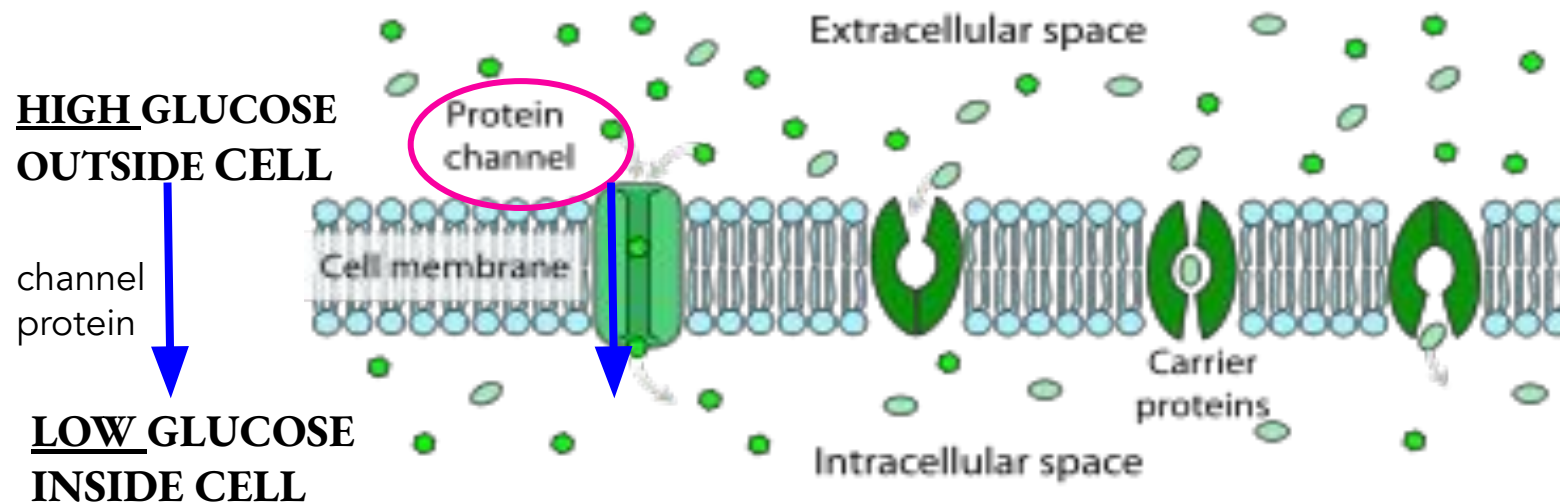
Passive Transport: Facilitated Diffusion

- **facilitated diffusion**- like simple diffusion, but uses the help of **transport proteins** (**channel or carrier**)
- Proteins aid molecules that normally could not pass or fit (“facilitate” their movement)
- Still moves from **HIGH → LOW** concentration
- Ex.) Glucose or amino acids moving into a cell



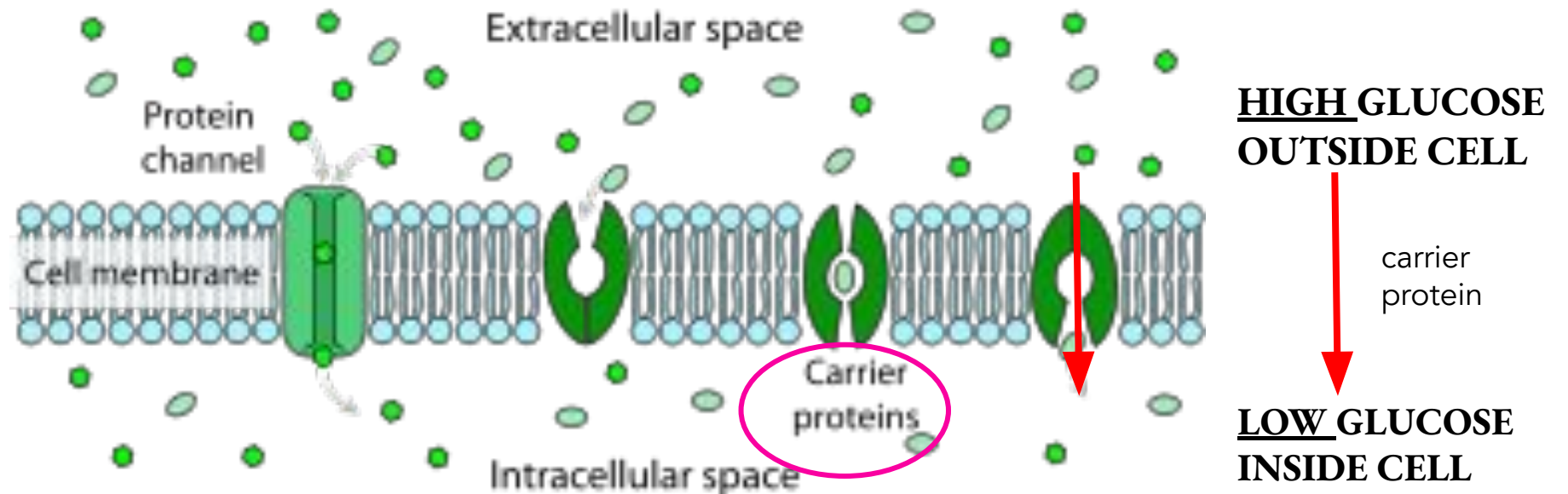
Facilitated Diffusion: Channel Proteins

- Channel proteins (type of **integral** protein) are **embedded** in the cell membrane
- Have **pore** that molecules can cross; molecules move randomly through these pores
- **aquaporins**- channel proteins that allow **water** to pass through more quickly than osmosis (simple diffusion)



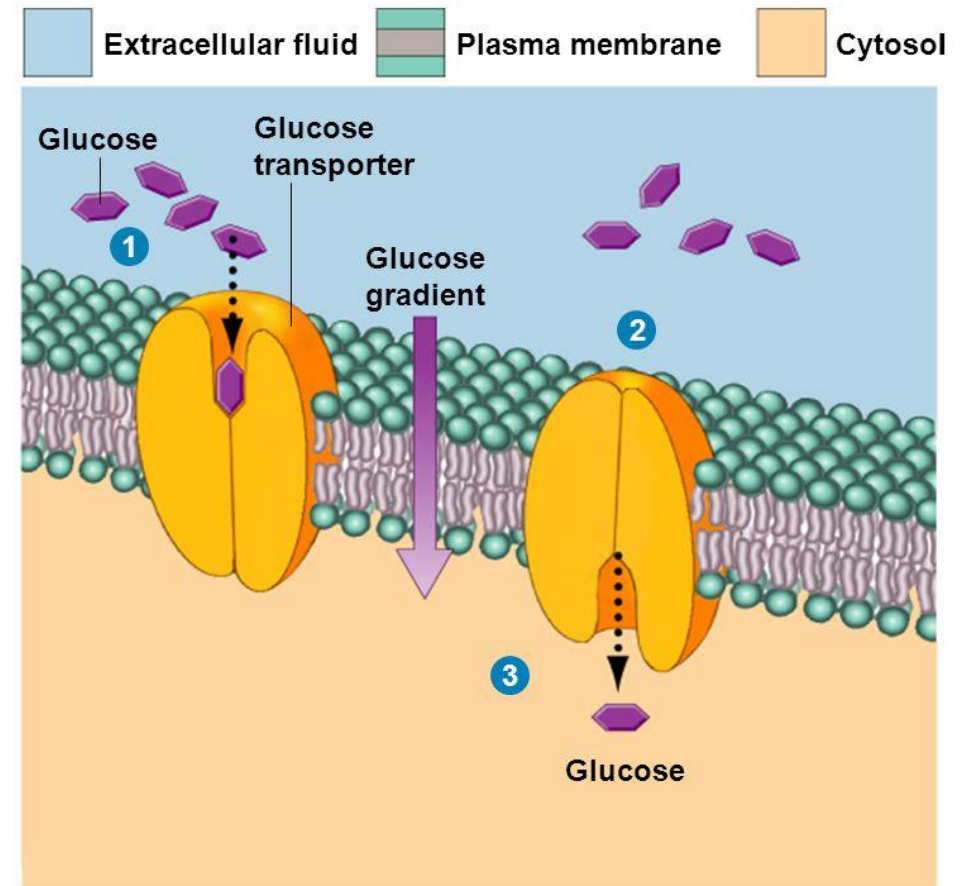
Facilitated Diffusion: Carrier Proteins

- **Carrier proteins** (type of **integral** protein) **change shape to move material** from one side to the other
- Some carrier proteins extend through the membrane while others do not

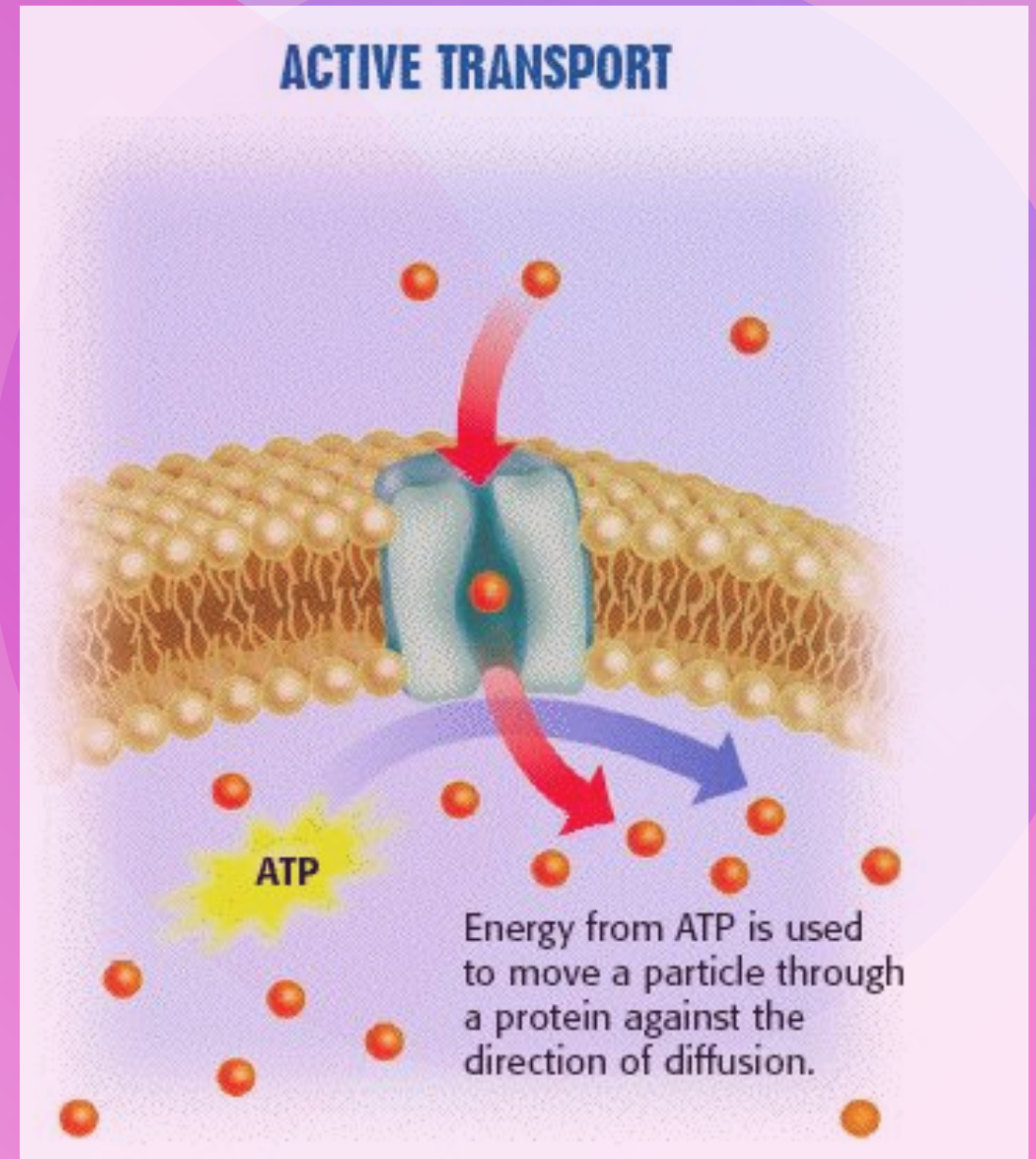


Facilitated Diffusion: Glucose

- **Glucose** is large enough that it needs help moving across the membrane
- moves from **high** → **low** glucose concentration using either a channel or carrier protein



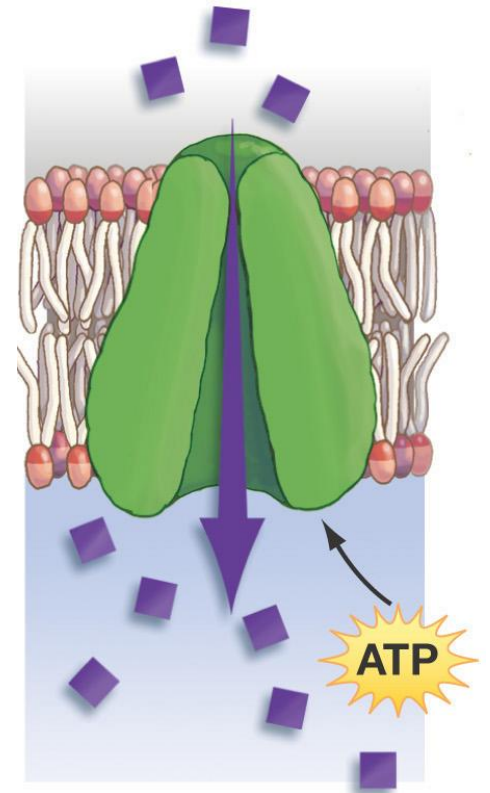
Active Transport



Active Transport

- active transport- processes that require energy (ATP)
- molecules move AGAINST or UP the concentration gradient (low $\rightarrow$ high concentration)

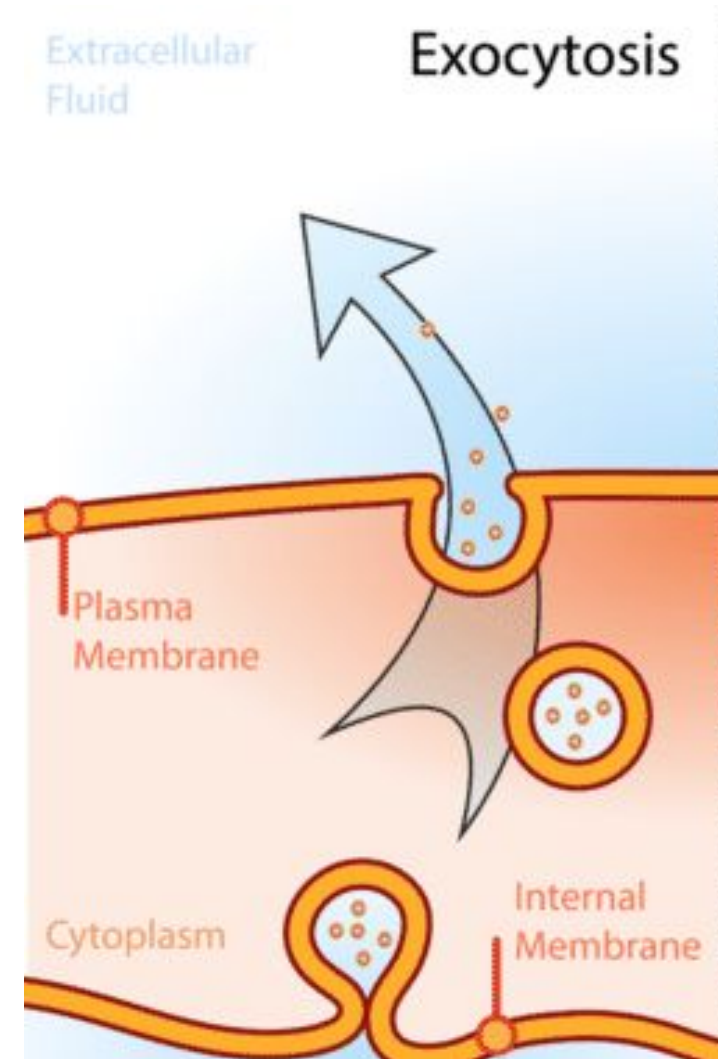
Active transport



Molecules again move through a transport protein, but now energy must be expended to move them against their concentration gradient.

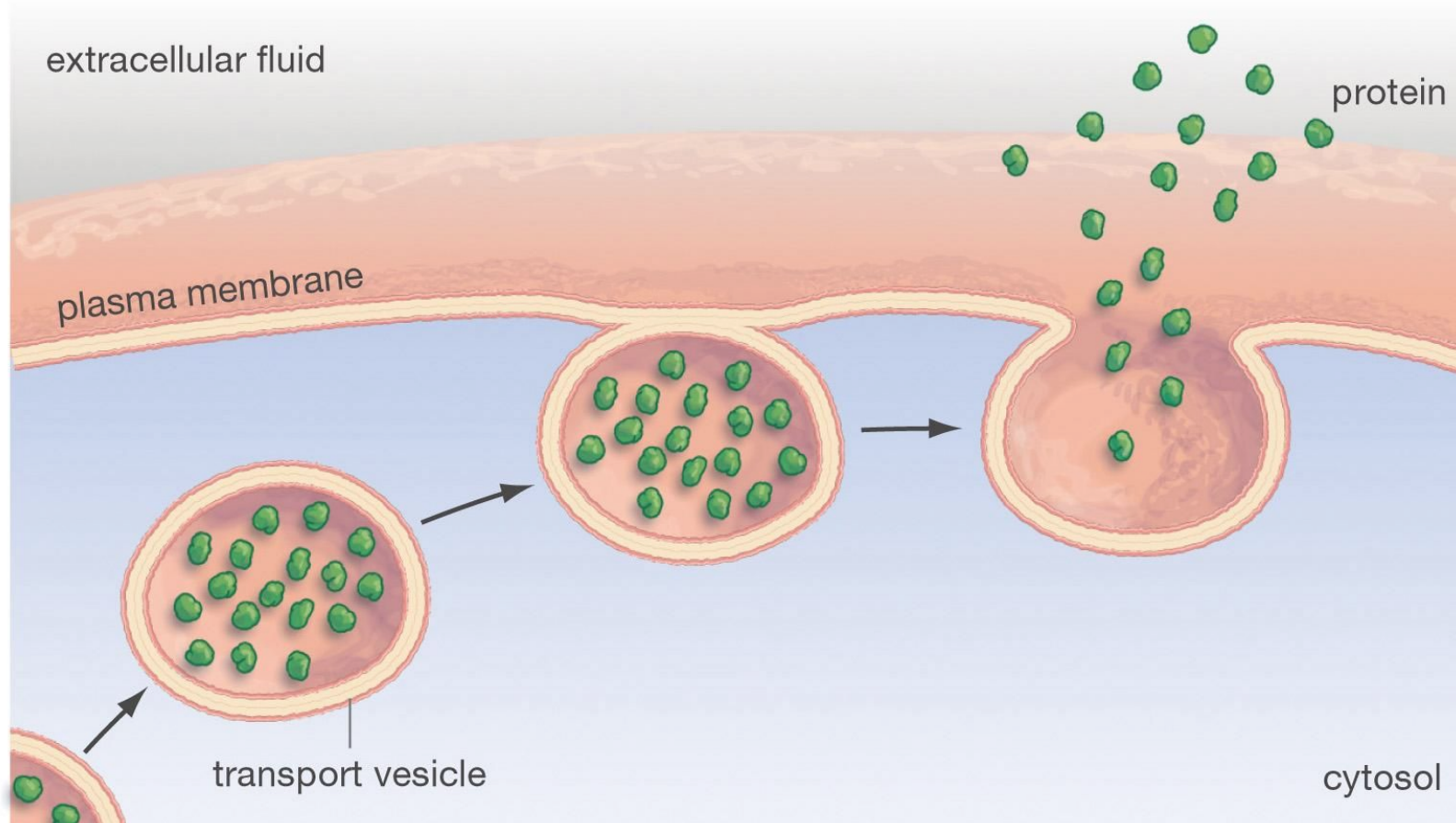
Active Transport: Exocytosis

- exocytosis- type of active transport where molecules are moved **in bulk** out of the cell using **vesicles** that fuse with the membrane
- requires energy (ATP)
- “exo” = “exiting” the cell
- Ex. **hormone secretion**, protein secretion, nerve cell communication

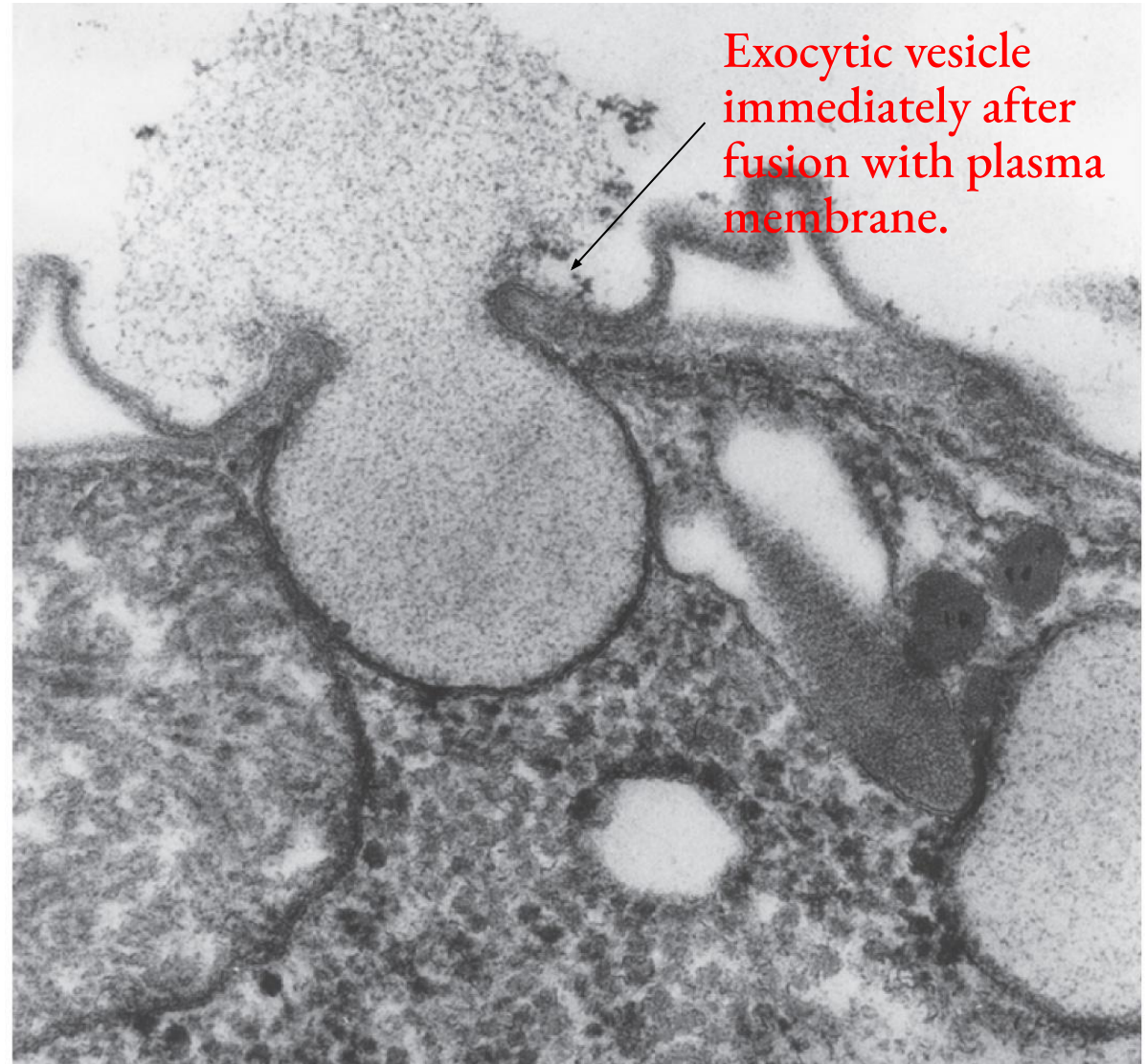


Exocytosis

(a) Exocytosis

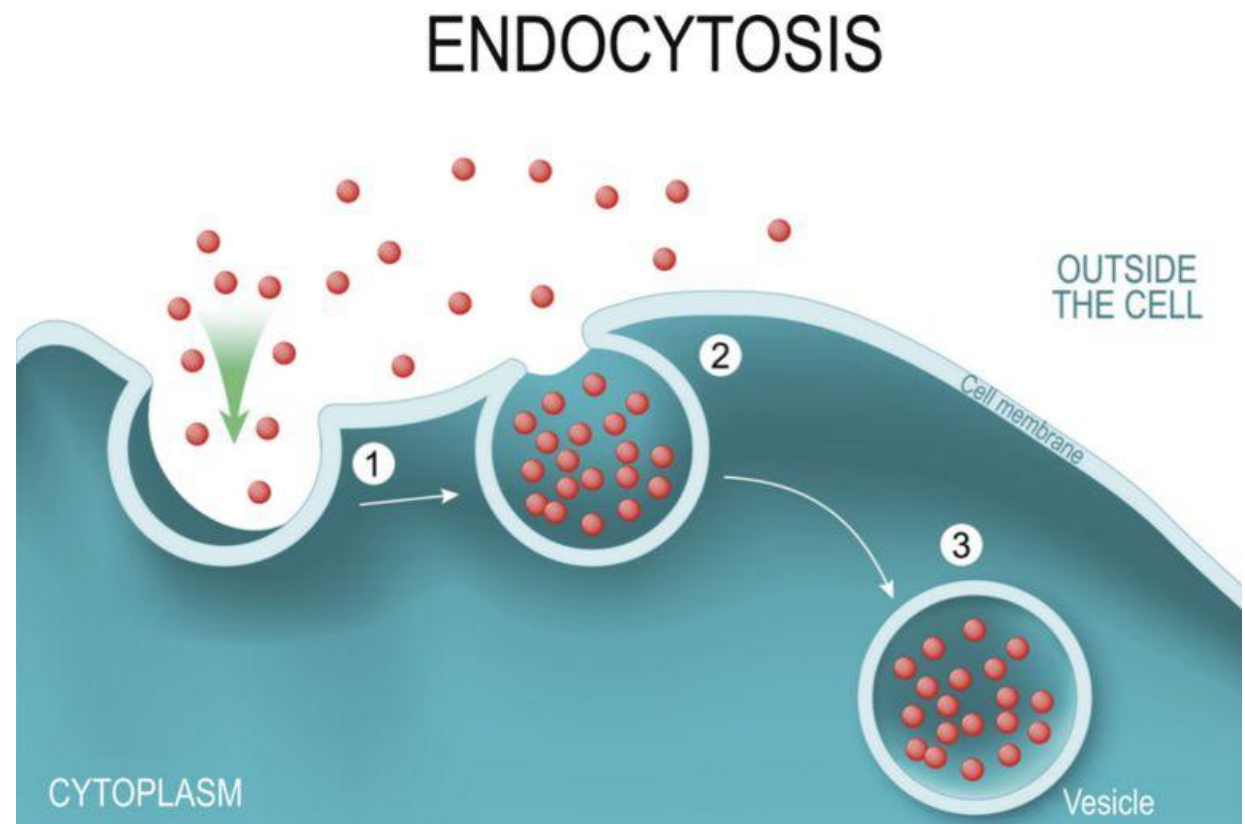


Exocytosis in the Cell

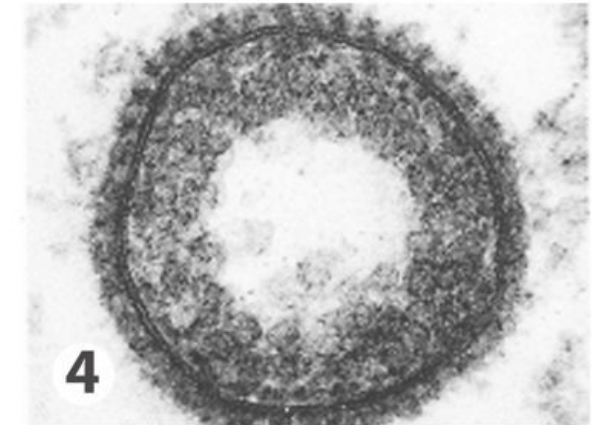
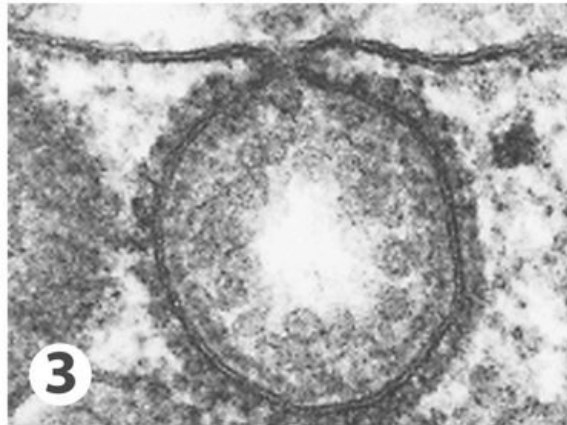
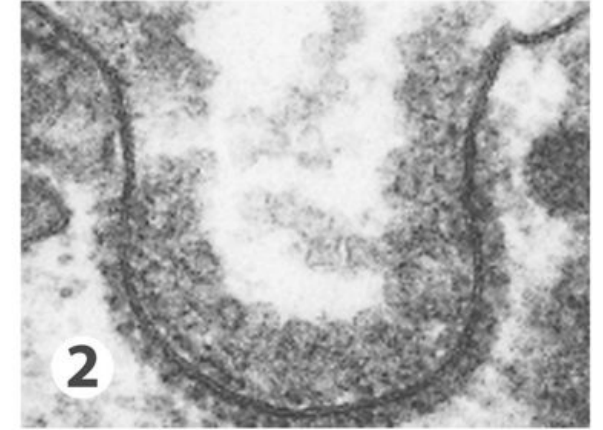
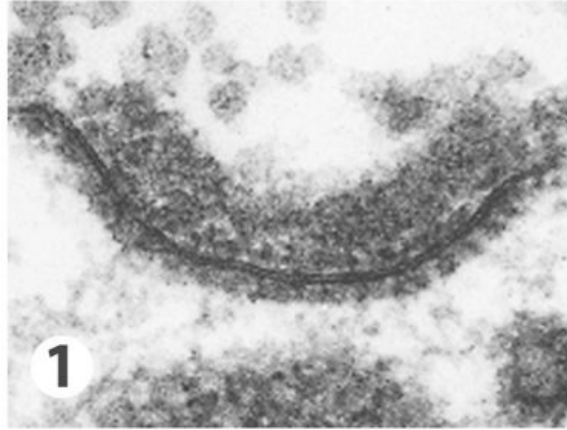


Active Transport: Endocytosis

- **endocytosis**- type of active transport where large molecules are moved **in bulk** **into the cell** using vesicles
- “**endo**” = **into** the cell
- 3 different types:
 - Pinocytosis (“cell-drinking”)
 - Phagocytosis (“cell-eating”)
 - Receptor-mediated

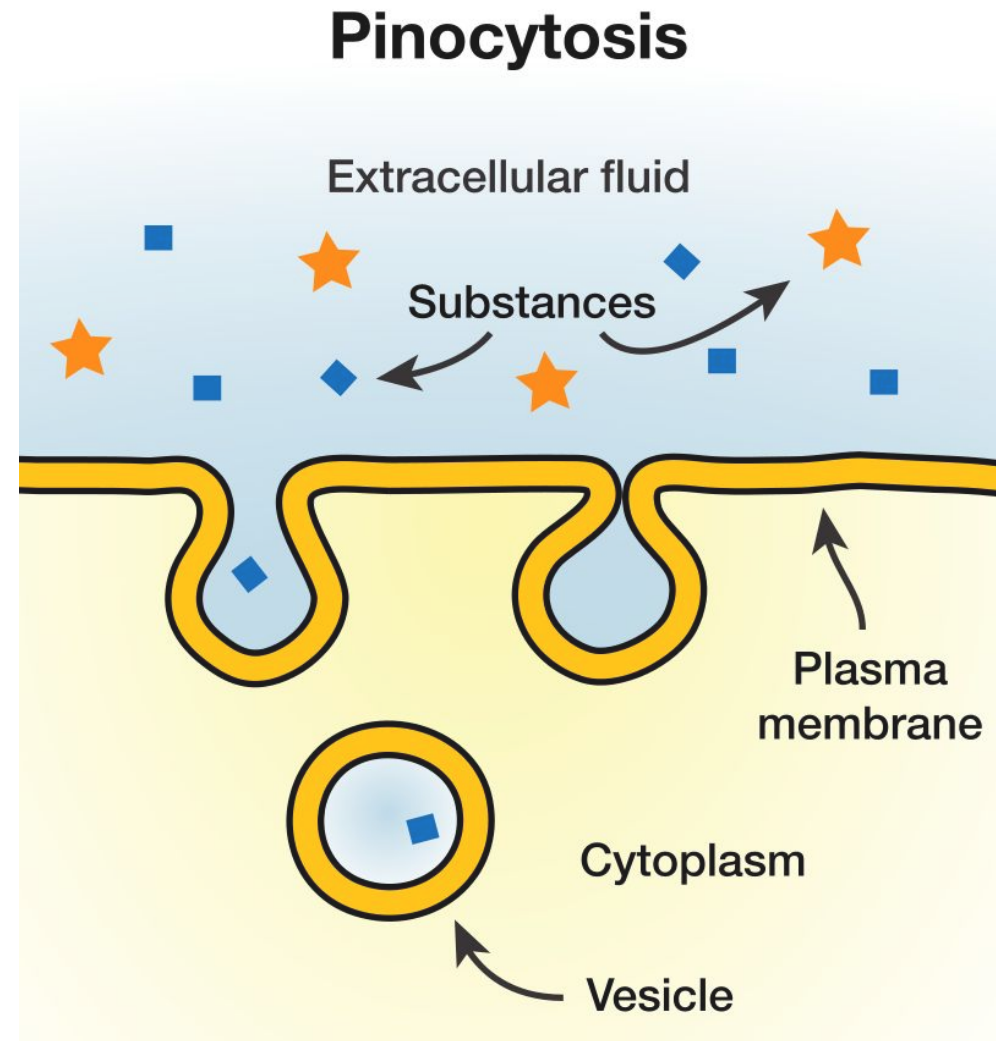


Endocytosis in the Cell



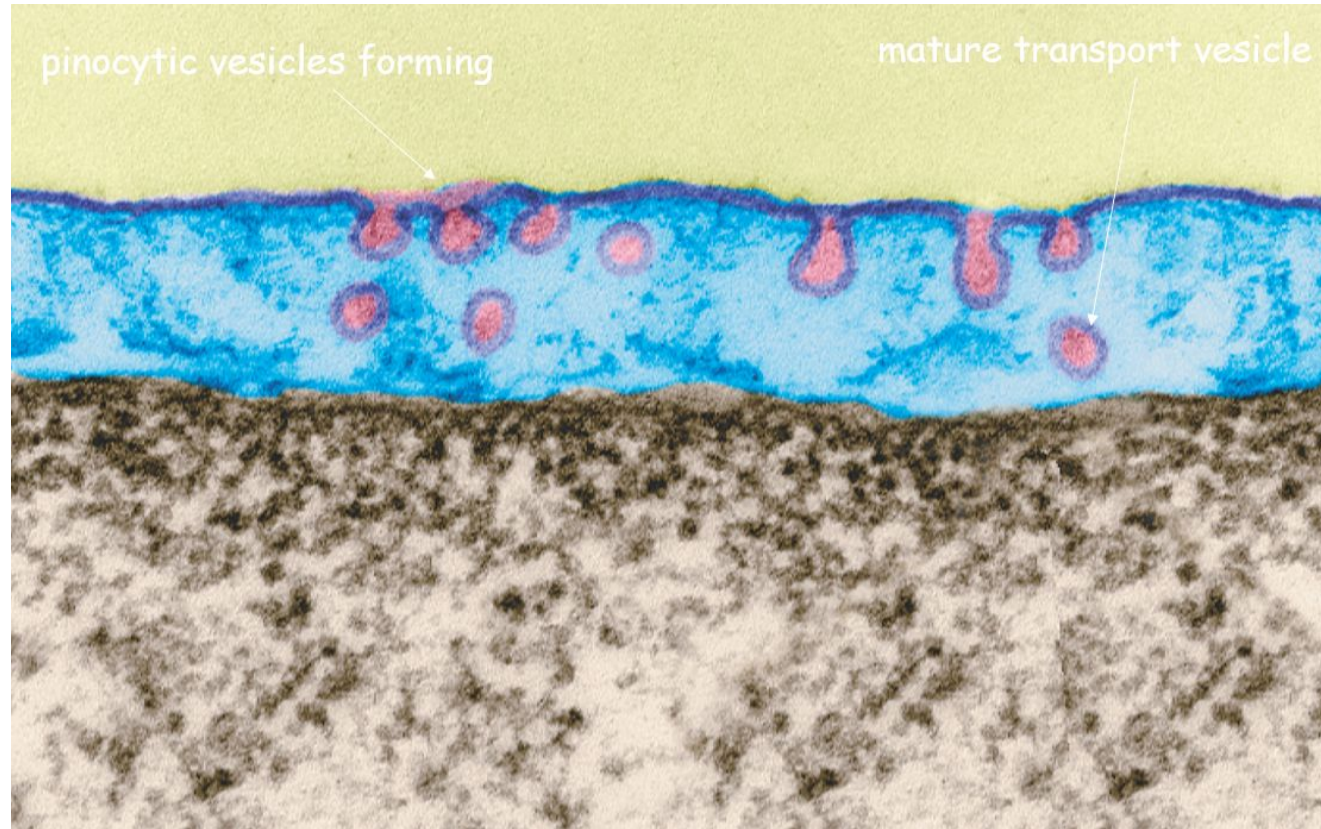
Endocytosis: **Pinocytosis**

- pinocytosis- “cell-drinking”
- cell takes in small amount of extracellular fluid with molecules
- Usually fats, vitamins, or waste molecules



Pinocytosis in the Cell

Transport
across a
capillary
cell (blue)



Outside cell
(extracellular)

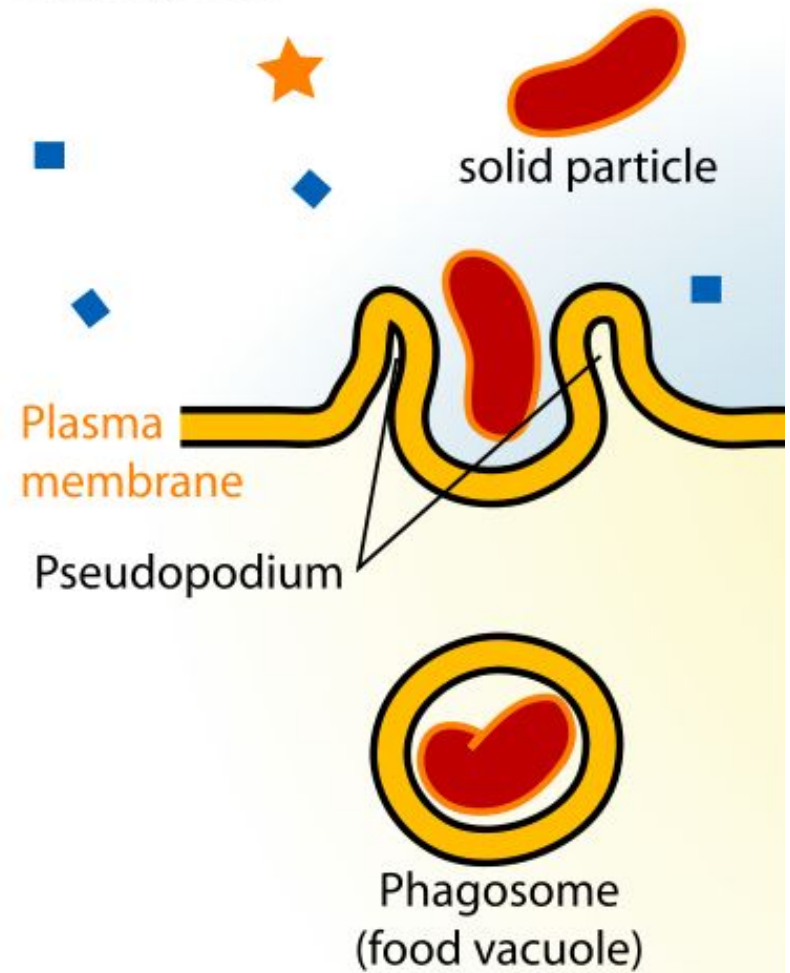
–*Cell Membrane*–

Inside cell
(Intracellular)

Endocytosis: Phagocytosis

- phagocytosis- “cell-eating”
- Cell engulfs large particles like **food or bacteria** into vesicles

Phagocytosis



Phagocytosis Examples

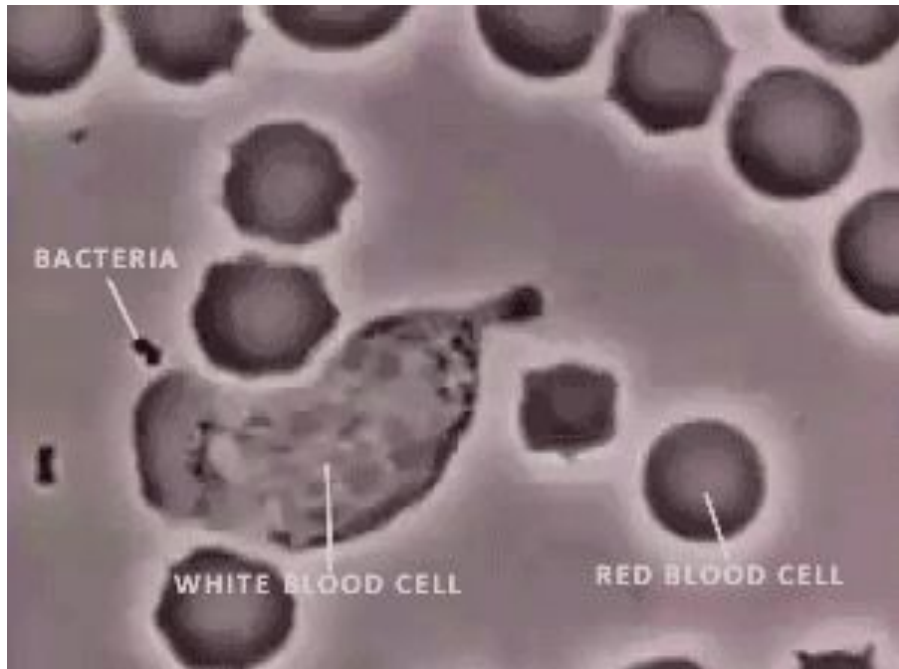
Macrophage engulfing bacteria



Immune cell (blue) engulfing yeast (yellow)



Phagocytosis Examples

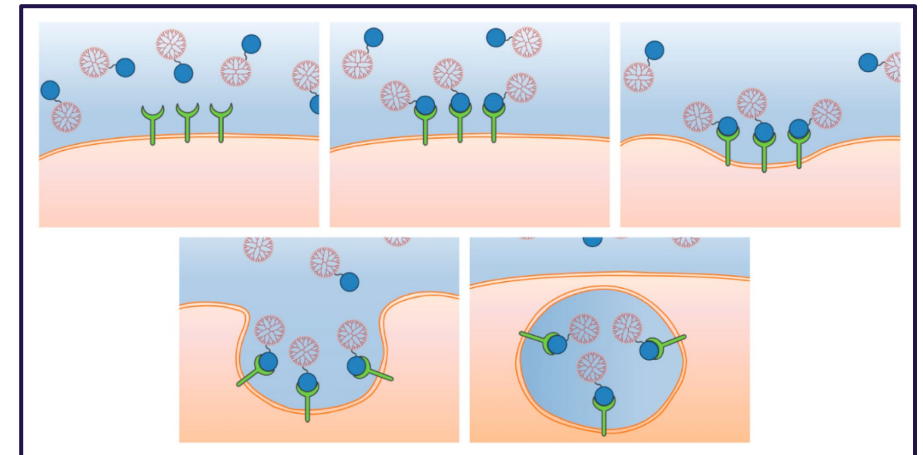
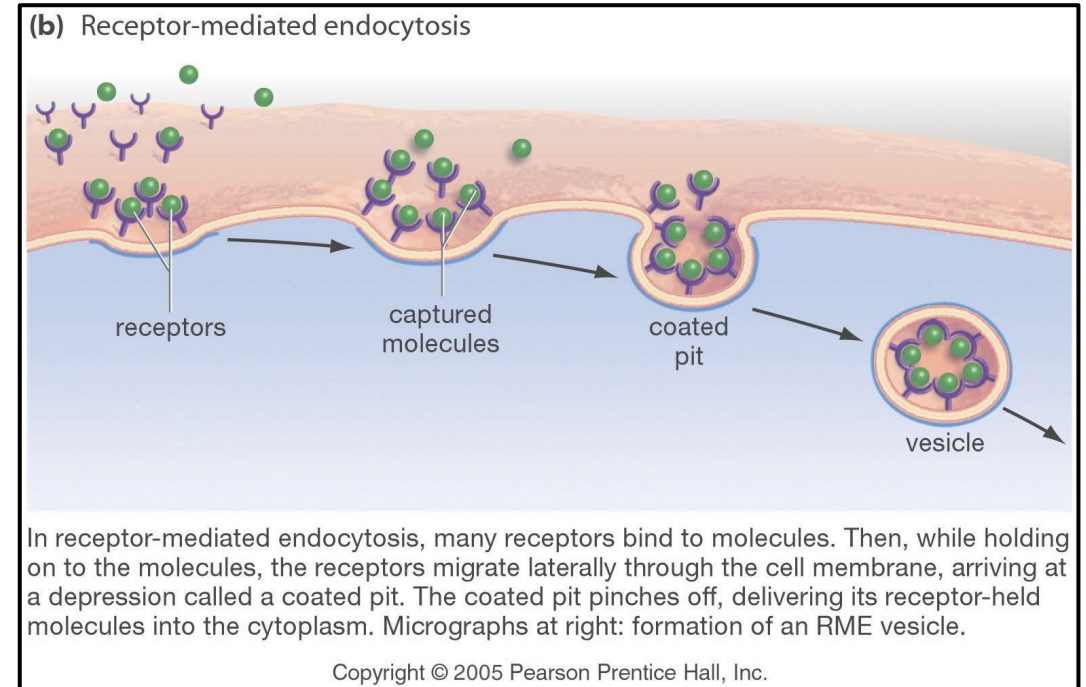


Amoeba eating paramecium

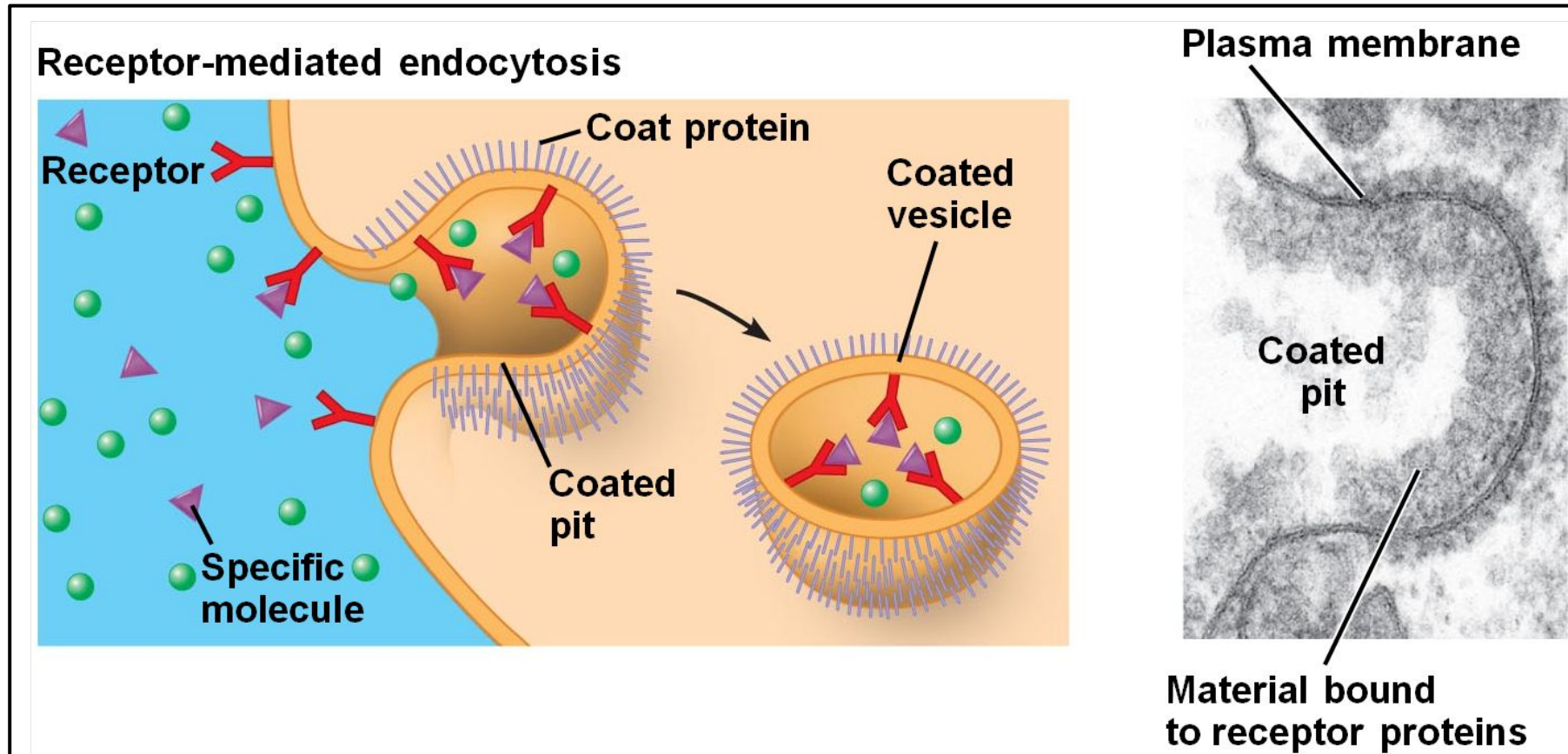


Endocytosis: Receptor-Mediated

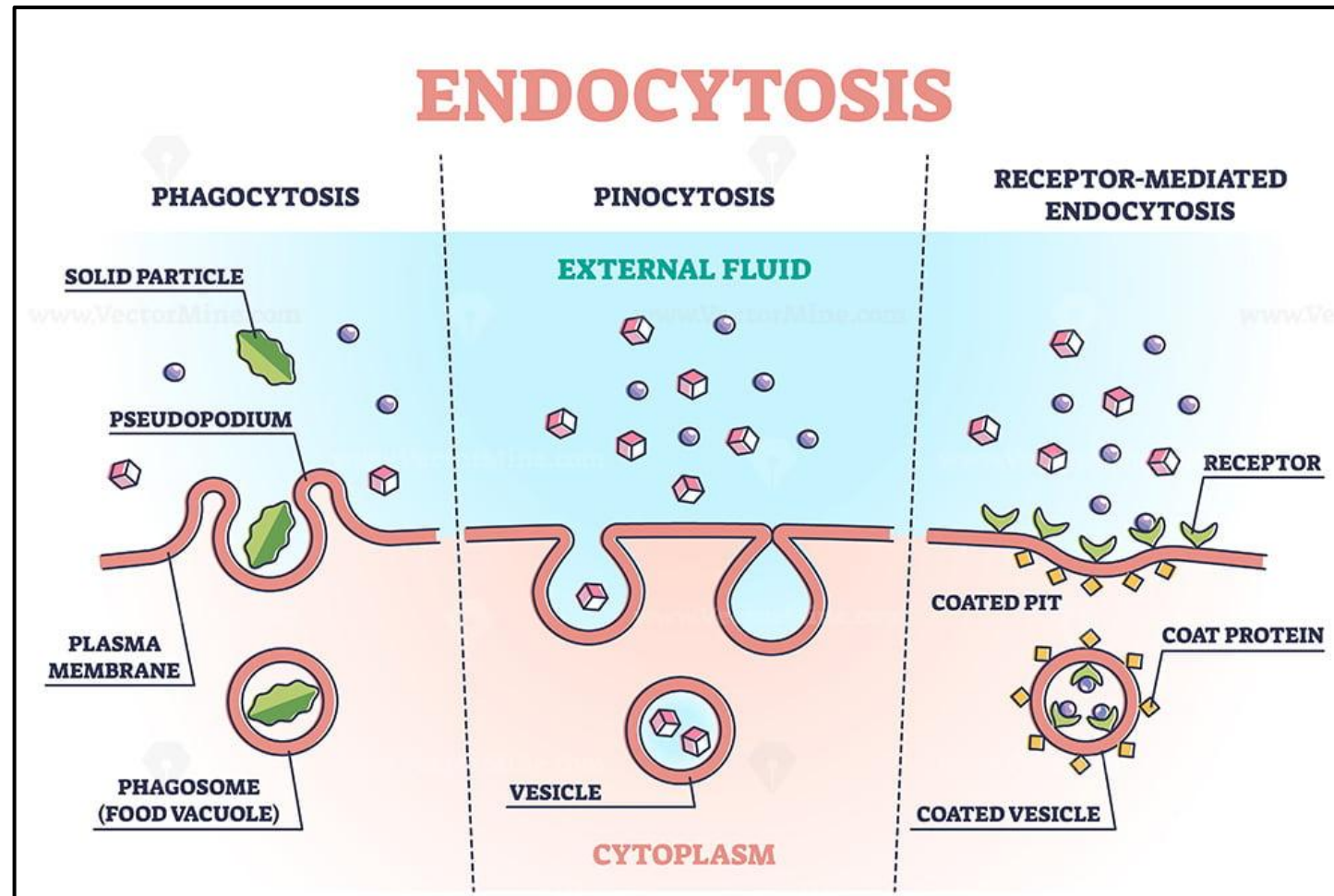
- receptor-mediated endocytosis-
receptor proteins (type of integral protein) bind to materials like hormones and cholesterol
- Materials are then engulfed by the cell when recognized



Receptor-Mediated Endocytosis

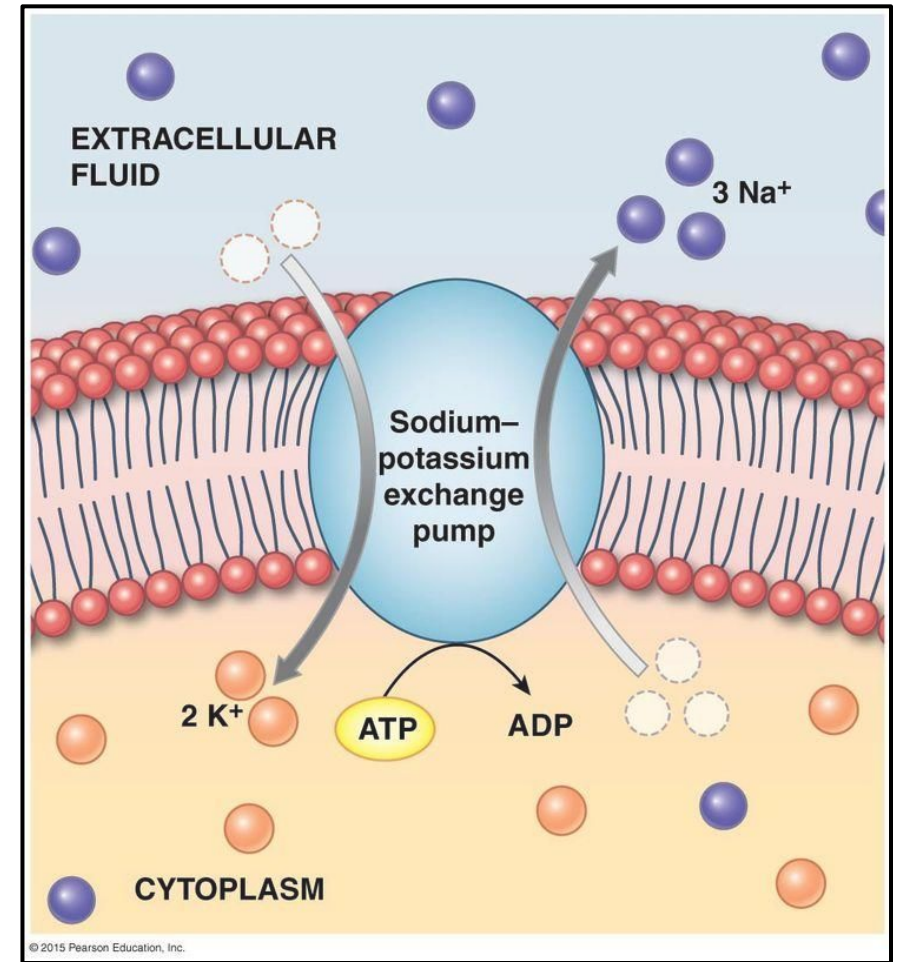


3 Types of Endocytosis (Active, INTO the cell)



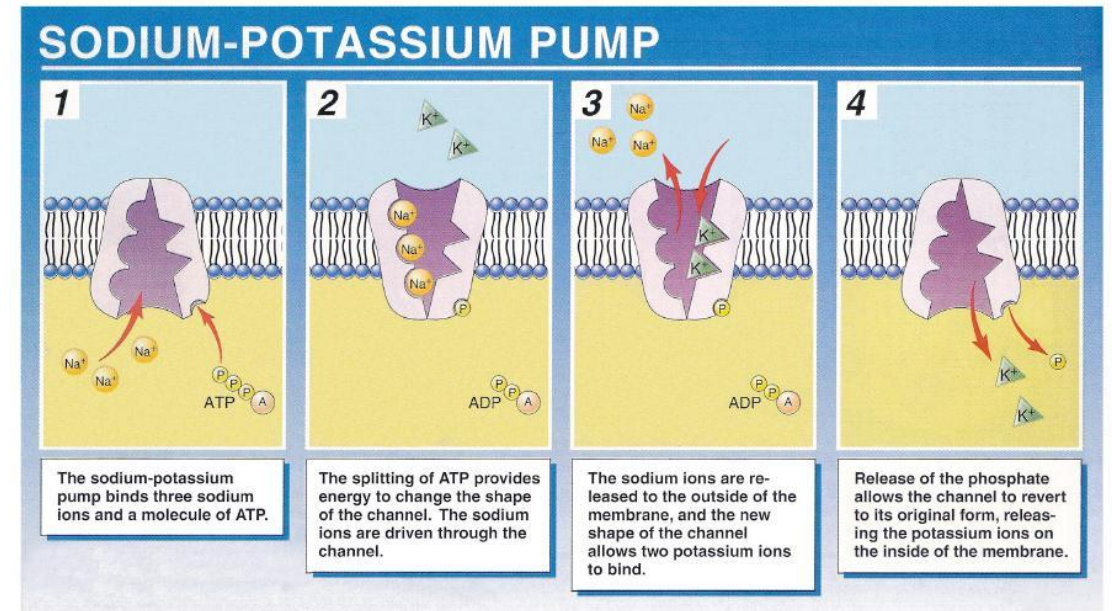
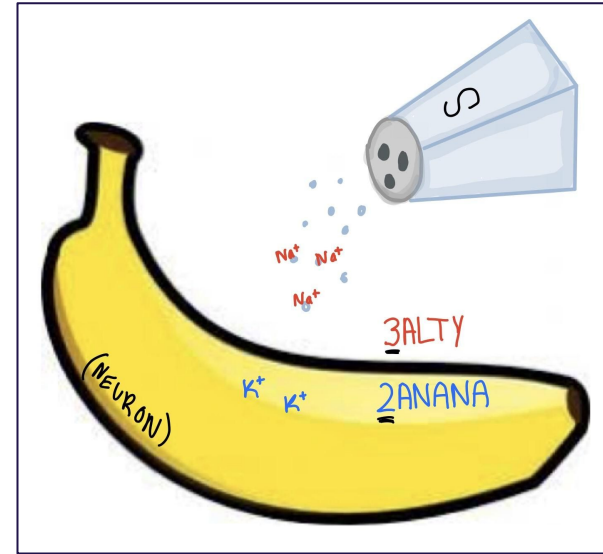
Active Transport: Sodium-Potassium Pump

- sodium-potassium pump- uses energy (ATP) to regulate sodium (Na^+) and potassium (K^+) concentrations
- a transport protein (carrier) changes shape to facilitate ion movement

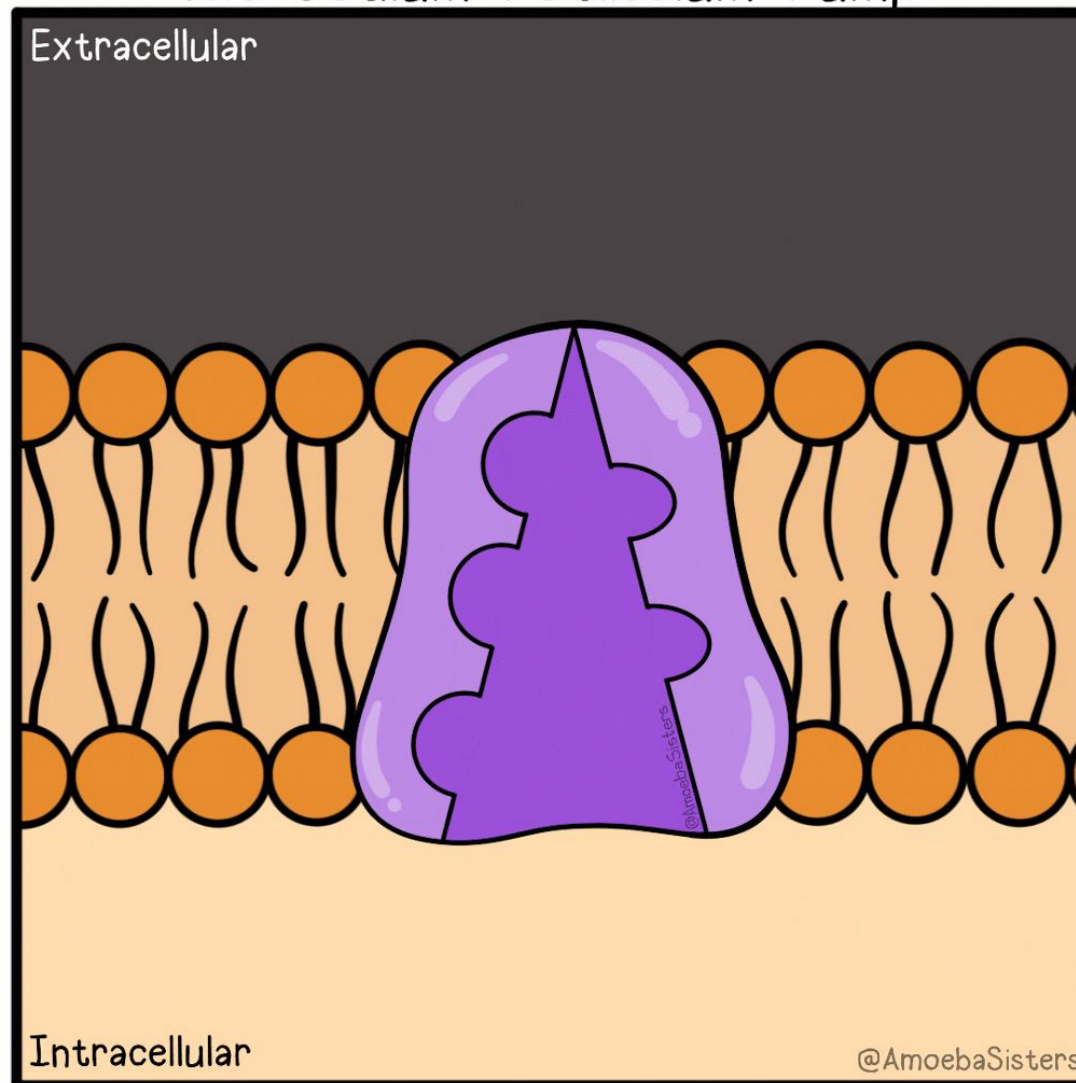


Sodium-Potassium Pump

- For every 2 potassium (K^+) ions pumped INTO the cell,
3 sodium (Na^+) ions are
pumped OUT
- makes the **inside** of cells have a **slightly negative charge**
- Cell is like a “salty banana”
(high Na^+ outside, high K^+ inside)



The Sodium Potassium Pump



Success Criteria - Passive Cell Transport (slides 18-34)

As you are studying, make sure you are able to do ALL of the following: (Make sure you recognize diagrams!)

- ❑ Describe in your own words what a concentration gradient is and what it means to be at equilibrium.
- ❑ Describe how molecules are moving in **passive cell transport** - **down and with** the concentration gradient (**high** → **low**) , or **up and against** the concentration gradient (**low** → **high**)?
- ❑ Compare & contrast **3 types** of passive transport: simple diffusion, osmosis, and facilitated diffusion.
- ❑ Set up and **solve diffusion and osmosis scenarios**, complete with arrows showing the movement of molecules and/or water (respectively). Be able to explain why molecules are moving in that direction.
- ❑ Describe the movement of water and how cells will look in **3 different solution types**: isotonic, hypertonic (plasmolysis), and hypotonic (cytolysis). Realize that these 3 terms are simply *adjectives* that describe how much solute is in something. (e.g, An egg in a hypotonic solution is hypertonic *compared* to the solution).
- ❑ Describe the similarities and differences between how animal and plant cells react when placed in hypertonic, isotonic, and hypotonic environments.
- ❑ Describe the types of molecules that undergo facilitated diffusion (large, polar, and/or charged) with the help of transport proteins (channel or carrier). An example you should be able to describe is **glucose**.

Success Criteria - Active Cell Transport (slides 35-52)

As you are studying, make sure you are able to do ALL of the following: (Make sure you recognize diagrams!)

- ❑ Compare & contrast **passive cell transport** vs. **active cell transport**. Address movement of molecules, the concentration gradient, energy, and proteins or cell structures (e.g. vesicles) involved.
- ❑ Compare & contrast the **2 examples of bulk-movement active transport**: exocytosis and endocytosis.
 - ❑ Describe the **3 types of endocytosis**: pinocytosis, phagocytosis, and receptor-mediated.
 - ❑ Explain how the **shape of receptor proteins** is essential for **receptor-mediated endocytosis** of molecules such as **hormones**.
- ❑ Describe the function of the **sodium-potassium pump** and how it contributes to homeostasis in cells. Be familiar with the relative concentrations of sodium and potassium inside and outside of the cell.